EXSU 620 / PROJECT 06 / DESIGN DOSSIER

Is the patient ventilating

Exhaled carbon dioxide and hypoventilation monitor. A bench prototype that confirms exhaled CO2, alarms on its loss, tells a disconnected line from apnoea, and measures and publishes its own response time.

PROJECT	06 · Is the patient ventilating
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COURSE	EXSU 620 Surgical Innovation 1 · Digital Prototyping · McGill University · Fall 2026
DEVICE	CO2M-6 bench prototype: Pico 2 W monitor in a printed case (firmware v0.6), laptop service, dashboard and phone messages
MEASURAND	Carbon dioxide concentration in gas sampled at the mouth (ppm)
SENSOR	Winsen MH-Z19C non-dispersive infrared module, 400 to 5,000 ppm (0.04 to 0.5 % CO2) version, read over UART once per second
REFERENCE	One-point zero against outdoor air (NOAA Mauna Loa, 426 ppm, September 2026). The simulation lab capnograph was not available before submission
REPOSITORY	github.com/azouzkm/CO2-Hypoventilation-Monitor
STATUS	Bench tested 1 to 4 Nov 2026. CR-1 met for presence; CR-2 met in 9 of 10 runs; CR-3 and CR-4 partly met, each with its measured cause and a designed fix. Not for patient use

> **How to read this dossier.** D1 states eight engineering requirements (ERs) traced to the card's four clinical requirements (CRs). D2 to D5 describe what was built, D6 and D7 how it was tested and what it achieved, D8 what can fail, and D9 the traceability matrix, the declared limitations and a one-page memorandum for a non-specialist reader. Every result names the ER it closes and the file it comes from.

SECTION 1

Clinical need and requirements, as issued

An unrecognised oesophageal intubation is fatal, and exhaled CO2 is the accepted proof that a tube is in the airway: the international airway consensus guideline treats exhaled CO2 as sustained only when it holds for at least seven breaths, with a peak more than 1 kPa (7.5 mmHg) above baseline [1]. During procedural sedation the CO2 signal shows respiratory depression before oxygen saturation falls: a median 35 s earlier than pulse oximetry in 101 sedated children [2], and a median 32 s from apnoea to a 4 % fall in saturation in 85 adults sedated for bronchoscopy [3].

The people who act on this device are the proceduralist and the sedation nurse, who stop the drug, reposition the airway and ventilate, and the anaesthetist called to a deteriorating sedation, who needs to know when it started (project card).

The team's literature review (`docs/D9-dossier/Literature Review.pdf` in the repository, 18 references) sets out this clinical background in more depth.

ID	REQUIREMENT, AS ISSUED
CR-1	The presence or absence of exhaled carbon dioxide shall be confirmed within a stated number of breaths and displayed unambiguously.
CR-2	Loss of carbon dioxide while the circuit is reported as connected shall raise an immediate alarm, since it is the signature of a misplaced tube or a disconnection.
CR-3	A blocked or disconnected sampling line shall be reported as a fault and shall never present as apnoea.
CR-4	Breathing rate derived from the carbon dioxide signal shall be recorded through the procedure, with alarms on a sustained fall.

SECTION 2, DELIVERABLE D1

Engineering requirements

Eight engineering requirements (ERs) turn CR-1 to CR-4 into numbers a bench test can pass or fail. Each names the clinical requirement it serves. The second table lists each one's range, sample rate, latency and accuracy, and the third gives the basis of every number.

What this hardware can be. The fitted sensor's 10–90 % rise is about 14 s, longer than one breath (4 s at 15 breaths/min), so the device gives a breath-averaged CO2 trend, not breath-by-breath capnography (ER-3, test 6). The ERs are written for a trend instrument and claim nothing more.

Bench conditions for ER-1 and ER-2 (test 5). Breath from a team member at the open end of a 0.5 m sampling line (3/16 in internal diameter vinyl), held 10 cm from the mouth. The line goes on the SAMPLE barb of the sealed printed flow-cell, and a 200 mL syringe on the VENT barb draws the sample at about 2 mL/s (120 mL/min). One sensor reading per second, firmware v0.6. D6 gives each other test's line, distance and draw.

ID	TRACES TO	REQUIREMENT	ACCEPTANCE CRITERION
ER-1	CR-1	The device shall show CO2 present (green) within 3 breaths of the first breath at the sampling line, at 15 breaths/min with the 0.5 m line.	Pass if green by the third breath in 10 of 10 scripted runs.

ID	TRACES TO	REQUIREMENT	ACCEPTANCE CRITERION
ER-2	CR-1, CR-2	The device shall raise the CRITICAL alarm (red, 10 pulses) within 30 s of breathing stopping while the line reads connected. The dashboard shall show it within 2 s, and every registered phone shall be messaged within 10 s of the box raising it.	Pass if red within 30 s of the stop cue in 10 of 10 scripted runs, and every critical alarm reaches the dashboard within 2 s and every phone within 10 s (notification log, all bench sessions).
ER-3	CR-1, CR-4	The device's response shall be measured and published: the delay from a breath to the first rise and to CO2 present, and the 10–90 % rise, for sampling lines of 0.5, 1 and 2 m at about 2 mL/s. Breath-by-breath capnography shall be claimed only if the 10–90 % rise is shorter than one breath (4 s at 15/min).	Pass if 3 single-breath trials per length are measured and published as mean and range, with the breath-by-breath decision stated.
ER-4	CR-1, CR-4	The device shall read CO2 from 400 to 5,000 ppm (0.04 to 0.5 %) once per second, within ±(50 ppm + 5 % of reading) of a reference after a one-point zero, with zero drift within ±50 ppm over 10 min in clean air.	Pass if the 10 min clean-air drift is within ±50 ppm, and every reference level (4 to 5 levels) is within the band.
ER-5	CR-3	A disconnected sampling line shall show as a technical fault (cyan, "Line off") within 2 s, and never as apnoea, including while the CO2 reading falls.	Pass if 10 of 10 line pulls show Line off within 2 s with 0 CRITICAL, and a 60 s disconnection while the CO2 drains raises 0 CRITICAL.
ER-6	CR-3	A blocked sampling line shall show as a technical fault within 30 s, and never as apnoea.	Pass if a 60 s pinch of the line during monitoring raises TECHNICAL within 30 s and 0 CRITICAL.
ER-7	CR-4	The device shall show and record the breathing rate derived from the CO2 signal, updated every 2 s and stored with every reading, from 5 to 20 breaths/min, within ±1 breath/min or ±10 % (the larger). It shall show "not resolved" rather than a number when the rhythm is unclear, and never a rate below 80 % of the true one.	Pass if, in every phase of the paced runs from 50 s in, a rate is shown at least 80 % of the time, its median is within tolerance, and no value is below 80 % of the paced rate, and every saved reading carries the rate or the reason it is withheld.
ER-8	CR-4	The device shall raise the ADVISORY alarm (amber, 3 pulses) within 90 s of breathing slowing below 8/min or to 60 % of its own rate, or within 60 s of the breath-averaged CO2 above room air falling 30 % below its own level. It shall return to NORMAL after 30 s back in range, and raise no advisory during steady breathing.	Pass if the advisory comes within these times in 10 of 10 paced slowing runs, and in none of 5 ten-minute steady runs.

2.1 Range, sample rate, latency and accuracy

ER	RANGE	SAMPLE RATE	LATENCY	ACCURACY OR TOLERANCE
ER-1	CO2 present or absent, at 15 breaths/min with the 0.5 m line	1 reading/s	≤ 3 breaths from the first breath	Present = 100 ppm above the 30 s low; 10 of 10 runs
ER-2	Breathing stopped while the line reads connected	1 reading/s; dashboard polls every 250 ms	Box ≤ 30 s; dashboard ≤ 2 s; phones ≤ 10 s	10 of 10 runs; every critical alarm delivered
ER-3	Lines of 0.5, 1 and 2 m at about 2 mL/s	1 reading/s	Measured and published: breath to first rise, to present, and the 10–90 % rise	Mean and range of 3 trials per length; breath-by-breath only if the 10–90 % rise is under 4 s
ER-4	400–5,000 ppm (0.04–0.5 %)	1 reading/s	Response is measured under ER-3	±(50 ppm + 5 % of reading) after a one-point zero; drift within ±50 ppm in 10 min
ER-5	Line on or off the SAMPLE barb	Clip read about every 20 ms, 0.3 s debounce	≤ 2 s	10 of 10 pulls; 0 CRITICAL, also while the CO2 falls
ER-6	Line pinched for 60 s during monitoring	1 reading/s	≤ 30 s	TECHNICAL, 0 CRITICAL
ER-7	5–20 breaths/min	Rate every 2 s from the last 64 s; stored with every reading	Judged from 50 s into each paced phase	±1/min or ±10 % (the larger); never below 80 % of the true rate
ER-8	Rate below 8/min or 60 % of its own; CO2 above room air down 30 % from its own level	Rate every 2 s; CO2 1 reading/s	≤ 90 s (rate), ≤ 60 s (CO2); NORMAL after 30 s back in range	10 of 10 slowing runs; 0 advisories in 5 ten-minute steady runs

2.2 Where the numbers come from

ER	NUMBERS	BASIS
ER-1	3 breaths at 15/min, 0.5 m line	The airway guideline judges sustained CO2 over at least 7 breaths [1], so the first sign of CO2 must come well inside that window. On 22 Oct the bare sensor rose 100 ppm above room air by breath 3 at 15/min, and the device's present rule uses that 100 ppm rise. The 0.5 m line holds about 9 mL, about 4.5 s of travel at 2 mL/s, or about one breath. 15/min is the middle of the normal adult range (12–20/min [6]); longer lines cost breaths, and ER-3 measures how many.

ER	NUMBERS	BASIS
ER-2	30 s, 2 s, 10 s	Apnoea is defined as more than 10 s without airflow [3], and the CO2 signal leads the oximeter by about half a minute (medians 32 and 35 s) [2, 3]. An alarm much later than 30 s would give that lead away. Absence is called from the falling trend, because the return to room air took 89 s on 22 Oct, and only after 12 s with no new rise; 30 s is about 7 breaths at 15/min. The dashboard repeats the box within two readings. 10 s bounds "at once" for a free messaging service (workshop: critical messages go immediately).
ER-3	0.5, 1 and 2 m; 4 s	The card asks the device to measure its own 10–90 % response and how the delay grows with line length, and to state which of the two it supports. A clinical sidestream monitor rises in up to 190 ms through a 2 m line [5]; one breath at 15/min lasts 4 s.
ER-4	400–5,000 ppm (0.04–0.5 %); ±(50 ppm + 5 %); ±50 ppm in 10 min	Range and accuracy are the fitted module's, from its label and manual [4]. A 50 ppm drift over 10 min is under 2 ppm in the 20 s the absence rule looks at, so drift alone cannot raise or clear an alarm. End-tidal CO2 of 35–45 mmHg is about 46,000–59,000 ppm (4.6–5.9 %), about ten times the top of this range (5,000 ppm, 0.5 %), so ER-4 states a working range and claims no breath values.
ER-5	2 s, 0 CRITICAL	Two readings plus the clip's 0.3 s debounce. "Never as apnoea" is CR-3's own wording; the hard case is a line pulled while the CO2 is falling.
ER-6	30 s, 0 CRITICAL	30 s matches ER-2's limit, so a blocked line is flagged no later than a stop would be; flagging it before CRITICAL needs the blockage sensed directly (D2).
ER-7	5–20/min; ±1/min or ±10 %; 80 %; every 2 s	Normal adult breathing is 12–20/min [6], and the range reaches below the 8/min advisory threshold. ±1/min is the resolution a rate is charted to. A falsely low rate would raise false advisories, so the device says "not resolved" rather than guess.
ER-8	8/min or 60 %; 30 %; 90 s and 60 s	8/min sits well below the normal adult range [6]. Sedation research counts a fall of more than 10 mmHg from baseline as hypoventilation [2], but the device's diluted level does not map onto end-tidal values, so 30 % is a working threshold, to be set against a reference instrument. The times follow the method: a rate needs 64 s of readings, and a fall must last 30 s to count as sustained (the card's 30 s back in range).

Three numbers depend on the bench rig and are declared with the results. ER-1's three breaths hold for the 0.5 m line; a 2 m line adds about 4 breaths (test 6). ER-4's accuracy band needs a reference instrument. ER-7 and ER-8 need breath to reach the sensor without saturating it.

SECTION 3, DELIVERABLE D2

System design

The box decides every patient alarm itself and keeps alarming if the laptop fails. The laptop records, shows and forwards what the box decided, and adds only what it alone can know: a lost link, and who acknowledged.

3.1 Architecture: three tiers, one alarm path

The design follows the course's architecture, in which the board keeps alarming if the laptop dies (workshop, slides 10 and 13).

1. **Device** (Raspberry Pi Pico 2 W on the Grove shield, CircuitPython 9.2.9, firmware v0.6). It reads the MH-Z19C once per second and decides presence or absence of CO2 (CR-1), the alarm state (CR-2, CR-3), and the breathing rate and advisory (CR-4). It shows the state on a 16-LED ring and a speaker, takes acknowledgements from a bedside button, and reads a safety clip on the sampling line. Each reading goes to the laptop as one text line and one JSON line.
2. **Laptop service** (Python, Flask). It reads the box over USB and saves every line to a raw log. It stores readings, alarms, acknowledgements and messages in SQLite, raises a technical alarm when no data arrives for 5 s, sends phone messages and serves the dashboard. It never decides a CO2 alarm of its own.
3. **Interface.** A browser dashboard on the laptop (localhost:8000) refreshes 4 times a second and uses no outside libraries, so it runs with no internet. It mirrors the ring and its sounds, plots CO2 and the breathing rate over 10 minutes, and takes acknowledgements with the user's initials. Course participants who opt in on the sign-up page get Telegram messages through CallMeBot, and their consent time is stored.

The USB cable that carries the data also powers the box, so a pulled cable turns the box off. The dashboard's "No data" alarm, then a phone message after a minute, take over (D8).

3.2 Sensing approach

Gas is sampled sidestream: drawn from the open end of a sampling line, through a sealed printed flow-cell around the sensor, and out of a VENT barb. The line (0.5 m of 3/16 in internal diameter vinyl for the scored runs; in test 6 also 1 m of the vinyl and the mask kit's detachable 2.1 m oxygen tubing) pushes onto the SAMPLE barb from outside the case, so no tubing sits inside the box. The cell's free volume is about 6.5 mL. On the bench a 200 mL syringe on VENT draws about 2 mL/s, or 120 mL/min (a clinical Microstream line samples at 50 mL/min [5]).

The sensor is a Winsen MH-Z19C non-dispersive infrared module. CO2 absorbs strongly at 4.26 µm, and the module reports its concentration over UART (9600 baud, read command 0x86) once per second, from 400 to 5,000 ppm, within ±(50 ppm + 5 % of reading), with a rated T90 under 120 s [4]. Its automatic self-calibration re-zeroes it to 400 ppm every 24 h, and holding its calibration (HD) wire to ground for 7 s re-zeroes it at once [4]. On 22 Oct that wire lay under the laptop's grounded case and held the reading at exactly 400 ppm; its tip is now insulated, and a reading stuck at 400 ppm is treated as a sensor fault.

Units. The device works in ppm, as the sensor reports. For comparison with clinical values, 1 % CO2 = 10,000 ppm ≈ 7.6 mmHg at sea level, and a normal end-tidal CO2 of 35–45 mmHg is about 46,000–59,000 ppm (4.6–5.9 %). The device's readings are concentrations of a diluted, breath-averaged sample, not end-tidal values.

Two properties of this sensor shape every other choice.

- **Range.** Its 5,000 ppm (0.5 %) ceiling is about a tenth of exhaled CO2, so the sample has to be diluted before it reaches the sensor. The fitted module is the 5,000 ppm version, and its manual documents no command to change the range [4]. On the bench the dilution came from holding the tube end 10 cm from the mouth; across the ten scored runs the peaks ranged from 2,228 to 5,000 ppm, and three runs reached the ceiling. At the ceiling the sensor cannot see breaths, so after 30 s at the top of the scale the box shows a technical fault, not green.
- **Speed.** Its 10–90 % rise is about 14 s (11–17 s), against 4 s per breath at 15/min, so breaths merge into one curve. Even the bare sensor in open air rose in 10.6–12.7 s (22 Oct), so the slowness is the sensor's own, not the line's. Presence is therefore called from a rise, absence from the falling trend, and the breathing rate from the small regular ripple that breathing leaves on the slope of the reading.

A draw is needed because a sealed cell without flow holds the last breath. On 25 Oct the cell read CO2 present with nobody breathing, and on 1 Nov, pinned at 5,000 ppm with nobody breathing, it stayed green for 3.5 minutes after the line went back on and raised CRITICAL only about 6 minutes after the last breath. The syringe gives a steady draw on the bench; the designed next revision replaces it with a pump at a fixed flow and a fixed room-air dilution (end of this section).

3.3 Alarm rules (on the box, firmware v0.6)

STATE	LIGHT AND SOUND	RAISED WHEN	CLEARED WHEN
STANDBY	one white light circling, silent	warm-up (75 s), before the first breath, or the button held 3 s	the first breath (CO2 present)
NORMAL	green, steady, silent	exhaled CO2 present	another state takes over
ADVISORY	amber, slow flash; 3 pulses every 20 s	the breathing rate below 8/min or 40 % below its own baseline, or the breath-averaged CO2 above room air 30 % below its baseline, for 30 s	30 s back in range; a press silences it for 2 min while the light stays
CRITICAL	red, fast flash; 10 pulses every 10 s; latched	CO2 lost while the line reads connected, or no CO2 within 30 s of the line or the sensor coming back	acknowledged (button or dashboard) and CO2 back, in either order; red and green in turn while CO2 is back but nobody has acknowledged
TECHNICAL	cyan, steady; a chirp every 30 s	sampling line off (safety clip open); the sensor not answering, at the bottom of its scale (400 ppm), or jumping faster than real CO2 can; the 3 min re-warm after a fault; 30 s at the top of the scale (5,000 ppm); no data for 5 s (laptop)	the fault ends

Presence and absence (CR-1). CO2 is present when the reading rises at least 100 ppm above its lowest point of the last 30 s. It is absent when the reading falls by at least 10 % of that rise, and at least 50 ppm, below its highest point of the last 20 s, or returns to within 25 % of the rise of where it started. Either call waits until 12 s have passed with no new rise of 25 ppm, so the dip between two breaths is not an absence. A slow fall (hypoventilation) is too gradual to trigger this; that is the advisory's job.

Priorities. A latched CRITICAL outranks TECHNICAL, which outranks ADVISORY and NORMAL. A fault never hides a patient alarm: a CRITICAL already latched stays red, with the technical problem shown beside it. A disconnected line is TECHNICAL, never CRITICAL, which is what makes CR-2's "while the circuit is reported as connected" hold. The pulse counts follow the pattern of IEC 60601-1-8 (10 pulses for high priority, 3 for medium); the device makes no claim to that standard.

Sensor faults. A one-second fall of more than 400 ppm, or rise of more than 1,200 ppm, is a sensor fault, not breathing. The fastest real changes seen through the cell were 163 ppm/s down and 654 ppm/s up, while the sensor glitches of 22 Oct jumped 1,006 to 4,600 ppm in one second. After a fault the sensor is trusted again only after a 3-minute re-warm, because on 22 Oct it overshot to 2,265 ppm while recovering.

Phone messages follow the workshop's rule. A critical alarm is sent at once, and again after 2 min to the backups too if nobody has acknowledged it; advisory and technical alarms are sent after 60 s; one follow-up goes out when an alarm someone received ends. Each message is tried up to three times (again after 10 s and 30 s), failures show on the dashboard, and each person gets at most 10 messages in 10 minutes. No patient identifier appears in any message.

Where the thresholds live. The alarm thresholds are in the firmware (`lib/cr1.py`, `lib/alarm.py`, `lib/cr4.py`), the only place a patient alarm is decided. The service takes its link timeout and message timing from `config.json` and mirrors what the box decided, so the box, the screen and the phones cannot disagree. The file's other thresholds, in percent, are left over from the starter code and are not read.

3.4 Data model

One SQLite file on the laptop, in write-ahead-log mode with a single writer. It holds the participants' contact details, so it stays out of the repository; the raw logs in `data/raw/` are the evidence for D7.

TABLE	ONE ROW PER	MAIN FIELDS
readings	reading (1 s)	time, ppm, CR-1 state, alarm state, ring display, acknowledged, quality, firmware, breathing rate and its reason, line on or off
alarm_events	alarm	priority, source (box or link), raised, cleared, reason, ppm at raise
acks	acknowledgement	alarm, time, who (initials), method (bedside or dashboard), confirmed by the box
device_events	banner line from the box	time, text
users	opted-in participant	name, role, channel, Telegram name or number, first list or backup, consent time, stop time, active
notifications	message	alarm, recipient, kind, text, attempts, queued, sent, status, error
calibrations	calibration fit	slope, intercept, reference, operator, residual

No patient identifiers enter any table or message.

3.5 Alternatives considered

OPTION CONSIDERED	DECISION	REASON AND EVIDENCE
Bare sensor in open air, breath blown at it	Rejected after testing	Anyone breathing nearby moved the reading: typing within reach raised it by 800–900 ppm, more than a deliberate breath (22 Oct). This led to the sealed flow-cell.
Sealed flow-cell with no draw, the breath pushing itself in	Rejected after testing	Our own first design, and the card's "patient's own breath" option. The cell held old breath: it read CO2 present with nobody breathing (25 Oct), and on 1 Nov, pinned at 5,000 ppm with nobody breathing, it stayed green for 3.5 min after the line went back on and raised CRITICAL only about 6 min after the last breath.
Kit non-rebreather mask as the sampling interface	Rejected after testing	With the line's end inside the mask the reading peaked near 1,200 ppm; minutes later, on the same rig, the bare tube end at the mouth reached 5,000 ppm within 25 s (2 Nov). The bench samples from the bare tube end.
Printed T-piece airway adapter (ISO 5356-1 22 mm and 15 mm cones)	Dropped (26 Oct)	The mask kit's connector is already a T, and once the mask was rejected the bench sampled from the bare tube end. The design returns as the first part of the next revision.
Syringe pulled in steps (35 mL every 10 s)	Rejected after testing	The reading sagged in each pause: three false CRITICAL alarms in 10 minutes (2 Nov).

OPTION CONSIDERED	DECISION	REASON AND EVIDENCE
Syringe pulled steadily (one 10 mL mark every 5 s, about 2 mL/s)	Selected for the bench	Not in the kit: a 200 mL MotoMaster syringe, bought because the kit has no pump. The 3/16 in vinyl tubing was bought with it, for the link to the syringe and the 0.5 m and 1 m lines; the kit's 2.1 m oxygen tubing was kept whole for test 6. No false CRITICAL or ADVISORY while breathing in the ten scored runs (test 5).
Electric pump (Adafruit 4700, 4.5 V)	Designed next step	Not in the kit and not available in time. It draws about 600 mA, too much to take from the box's USB supply, and is not rated for continuous use [7], so the next revision needs a continuous-duty pump on its own supply.
Alarm decided on the laptop	Rejected	A laptop or service failure would silence it. With the service stopped mid-alarm, the box kept flashing and sounding (test 7).
CO2 reading used to detect a blocked line	Rejected after testing	During a 60 s pinch the reading fell and CRITICAL came 35 s after the pinch, as in an apnoea (test 4), so CO2 alone cannot tell the two apart. Designed fix: sense the line pressure or the pump's current (below).
Breath peaks counted one by one on the box	Rejected	Each breath leaves only about 2 ppm of ripple on the reading (22 Oct), so single peaks are easily missed, and every missed peak lowers the count: a falsely low rate raises a false slow-breathing advisory. The spectral method withholds the rate when in doubt; peak counting stays as an offline cross-check (`tools/peak_rate.py`).
WhatsApp through CallMeBot	Replaced by Telegram	CallMeBot's WhatsApp bot was full when tried (27 Oct and 1 Nov); its Telegram bot is the same free service from the card's list.
FastAPI service, the team's first skeleton	Replaced by Flask	Both are on the card's list. Flask needs one package instead of three (FastAPI, Uvicorn, Pydantic), and one laptop serving one browser needs nothing more. The skeleton's REST contract (`/api/latest`, `/api/ack`), its SQLite file in WAL mode and its table plan were kept.
Chart.js or Plotly on the dashboard	Rejected	The trend is drawn on a plain canvas, so the dashboard works with no internet and nothing can change under it before a demonstration.
Frosted dome over the ring (card list)	Not in the kit	A printed PLA light cap diffuses the ring instead.

3.6 Designed next revision (not built)

The bench results point to one change that addresses three failures at once: a fixed flow with a fixed dilution.

1. A T-piece at the mouth or airway, with the sample taken from its side port.
2. An airline control valve and a tee open to room air, set once so that steady breathing reads 2,500–3,000 ppm, well below the ceiling, then taped.

3. The 0.5 m sampling line with a water trap, then the sensor cell.
4. A continuous-duty pump on VENT at a fixed flow, on its own supply.

This takes the operator out of the flow (D8), keeps breath below the 5,000 ppm ceiling that delayed one alarm and hid the breathing rate (ER-2, ER-7), and gives a blocked line a signature of its own: the line pressure, or the pump's current, changes when the line is blocked, while an apnoea leaves them unchanged (ER-6). A 10,000 ppm version of the module would add headroom; breath-by-breath capnography would need a faster NDIR module with a range up to 100,000–200,000 ppm (10–20 %).

SECTION 4, DELIVERABLE D3

Component, module and system chart

The monitor is four modules of 23 components, joined by 12 named interfaces. Every patient alarm is decided in one module, the device firmware (M2); the laptop service (M3) and the interface (M4) record, show and forward it.

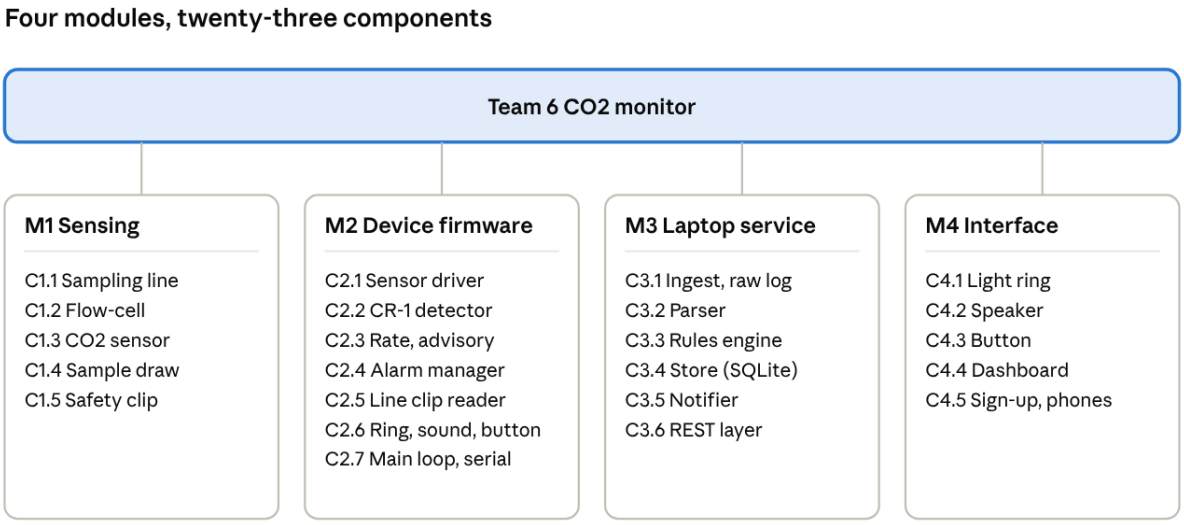


Figure 1. Component, module and system hierarchy: four modules and 23 components. Each component has one row in the components table, and each connection between them one row in the interface table.

4.1 Components

MODULE	COMPONENT	WHAT IT IS	WHERE
M1 Sensing	C1.1 Sampling line	0.5 m of 3/16 in (4.8 mm) internal diameter clear vinyl for the scored runs, about 9 mL; its open end is held 10 cm from the mouth. Test 6 also used 1 m of the vinyl and the mask kit's detachable 2.1 m oxygen tubing	Vinyl bought; 2.1 m tubing from the kit
M1 Sensing	C1.2 Flow-cell	Printed tray and lid sealed around the sensor (lid seam and wire exit taped), about 6.5 mL of free volume. It stands on edge inside the printed case, its two barbs (5.6 mm ridges as drawn, 5.3 mm measured) through the front panel as SAMPLE and VENT	`hardware/cad/`
M1 Sensing	C1.3 CO2 sensor	Winsen MH-Z19C NDIR module, 400–5,000 ppm (0.04–0.5 %) version, 5 V supply, UART output. Its calibration (HD) wire is insulated and unused	Kit
M1 Sensing	C1.4 Sample draw	200 mL MotoMaster syringe on VENT through 30 cm of the same tubing, pulled at about 2 mL/s (one 10 mL mark every 5 s), emptied off the line, never back through the cell	Bought
M1 Sensing	C1.5 Safety clip	Break-wire loop from GP21 to GND: a plug taped along the line sits in a socket beside SAMPLE and pulls out when the line comes off	`hardware/wiring.md`, `docs/photos/`
M2 Device firmware	C2.1 Sensor driver	Reads CO2 once per second (command 0x86), with the two data wires either way round; keeps the ring, sound and button running while it waits up to 0.5 s for a reply	`firmware/lib/mhz19c.py`

MODULE	COMPONENT	WHAT IT IS	WHERE
M2 Device firmware	C2.2 CR-1 detector	Present: a rise of 100 ppm above the 30 s low. Absent: a fall of 10 % of that rise (at least 50 ppm) below the 20 s high, or back within 25 % of the start, after 12 s with no new rise. Sensor faults: no answer, stuck at 400 ppm, impossible jumps; 3 min re-warm	`firmware/lib/cr1.py`
M2 Device firmware	C2.3 Breathing rate and advisory	Rate every 2 s from the spectrum of the reading's slope over 64 s, shown only when the rhythm passes its checks; advisory on a sustained fall in rate or level	`firmware/lib/cr4.py`
M2 Device firmware	C2.4 Alarm manager	STANDBY and the card's four alarm states (D2): latching, acknowledgement, priorities, silence for 2 min	`firmware/lib/alarm.py`
M2 Device firmware	C2.5 Line clip reader	Safety-clip state, accepted after 0.3 s	`firmware/lib/lineclip.py`
M2 Device firmware	C2.6 Indicator and annunciator	Ring colour and pattern per state; alarm tones in the IEC 60601-1-8 pulse pattern; button press and 3 s hold	`firmware/lib/indicator.py`, `firmware/lib/annunciator.py`
M2 Device firmware	C2.7 Main loop and serial writer	One reading per second; one text line and one JSON line to the laptop; takes `ACK` from the laptop	`firmware/code.py`
M3 Laptop service	C3.1 Serial ingest and raw log	Finds the Pico's port, again after a replug, and saves every line to `data/raw/`	`service/logger.py`
M3 Laptop service	C3.2 Parser	Text and JSON lines to readings, with a quality flag	`service/parse.py`
M3 Laptop service	C3.3 Rules engine	Mirrors the box's alarm; raises the link-lost technical alarm after 5 s without data; records acknowledgements	`service/rules_engine.py`
M3 Laptop service	C3.4 Store	The seven SQLite tables of D2	`service/store.py`
M3 Laptop service	C3.5 Notifier	Who gets which message when; retries; at most 10 messages per person in 10 min	`service/notifier.py`
M3 Laptop service	C3.6 REST layer	Flask app serving both pages and the JSON API; reads `config.json`	`service/main.py`, `service/core.py`
M4 Interface	C4.1 Light ring	NeoPixel Ring 16 (SK6812 LEDs) at 8 % brightness under a printed PLA light cap, seen through a window in the case lid	Kit; cap printed
M4 Interface	C4.2 Speaker	8 Ω speaker on GP18 through a 220–470 Ω resistor. The kit's PAM8302A amplifier was not soldered, so the tone is quiet	Kit
M4 Interface	C4.3 Acknowledge button	16 mm panel button on GP20, on the front panel ("HOLD 3 s = STOP")	Kit
M4 Interface	C4.4 Dashboard	Browser page: ring and alarm band, CO2, breathing rate, line state, two 10-minute charts, alarm sounds, acknowledge with initials	`dashboard/index.html`
M4 Interface	C4.5 Sign-up page and phones	Opt-in page that stores the consent time; Telegram messages through CallMeBot	`dashboard/register.html`

4.2 Interfaces

ID	DIRECTION	SIGNAL AND PROTOCOL	RANGE AND RATE
IF-1	Mouth → sampling line → flow-cell → VENT → syringe	Gas, drawn sidestream by the syringe	About 2 mL/s (120 mL/min). Breath is about 40,000–60,000 ppm (4–6 %) at the mouth; the sensor reads 400–5,000 ppm (0.04–0.5 %)

ID	DIRECTION	SIGNAL AND PROTOCOL	RANGE AND RATE
IF-2	Pico ↔ CO2 sensor	UART 9600 baud 8N1, 3.3 V logic: read command 0x86 out on GP0, 9-byte reply in on GP1. Sensor supply 5 V (5.18 V measured)	One reading per second; 400–5,000 ppm in 1 ppm steps
IF-3	Pico → light ring	Single-wire addressable LED data (800 kHz) on GP16, the Grove shield's D16 socket; 5 V supply	16 LEDs at 8 % brightness (about 0.1 A); one colour and pattern per state
IF-4	Pico → speaker	Square-wave tones on GP18 through the resistor	1.6–3.0 kHz. CRITICAL: 10 pulses at 2.4 kHz every 10 s; ADVISORY: 3 pulses at 1.8 kHz every 20 s; TECHNICAL: a 3.0/2.4 kHz chirp every 30 s
IF-5	Button → Pico	GP20 to GND, internal pull-up, active low, 40 ms debounce	Press = acknowledge (sound off for 2 min); hold 3 s = stop monitoring (STANDBY)
IF-6	Pico ↔ laptop service	USB CDC serial, opened at 115,200 baud. To the laptop: one text line and one JSON object per reading. To the board: `ACK`	One reading per second: ppm 400–5,000 or empty (no answer); alarm one of the five states; rate 5–24/min or empty, with its reason; line on or off
IF-7	Service → database	SQLite file in write-ahead-log mode, one writer	One row per reading, plus alarm, acknowledgement and message rows
IF-8	Dashboard and sign-up page ↔ service	HTTP on localhost:8000, JSON REST: `GET /api/latest`, `/api/readings`, `/api/alarms`, `/api/notifications`, /api/health, /api/users; `POST /api/ack`, `/api/users`	Dashboard polls `/api/latest` every 250 ms (4 times a second) and the alarm log every 3 s; an acknowledgement carries the user's initials; the sign-up page also sends test, stop and start requests for each participant
IF-9	Service → CallMeBot → phones	HTTPS GET per message to CallMeBot's Telegram API; retried after 10 s and 30 s	CRITICAL at once, again at 2 min to the backups if nobody has acknowledged; ADVISORY and TECHNICAL after 60 s; at most 10 per person in 10 min
IF-10	Safety clip → Pico	GP21 to GND through the clip, internal pull-up: low = line on, high = line off; a broken wire also reads off	Read about every 20 ms; a change counts after 0.3 s
IF-11	Laptop USB → device	5 V USB power for the Pico, sensor and ring, on the same cable as IF-6	5.18 V at the Pico's VBUS; pulling the cable powers the device off
IF-12	Laptop ↔ board's CIRCUITPY drive	USB mass storage, on the same cable: firmware files copied onto the drive; boot_out.txt read back	Any file change restarts the board: STANDBY and a 75 s warm-up

SECTION 5, DELIVERABLE D4

System diagrams

Four diagrams: what connects to what, how the alarm states change, how the device is wired, and how one reading becomes a phone message.

5.1 Block diagram

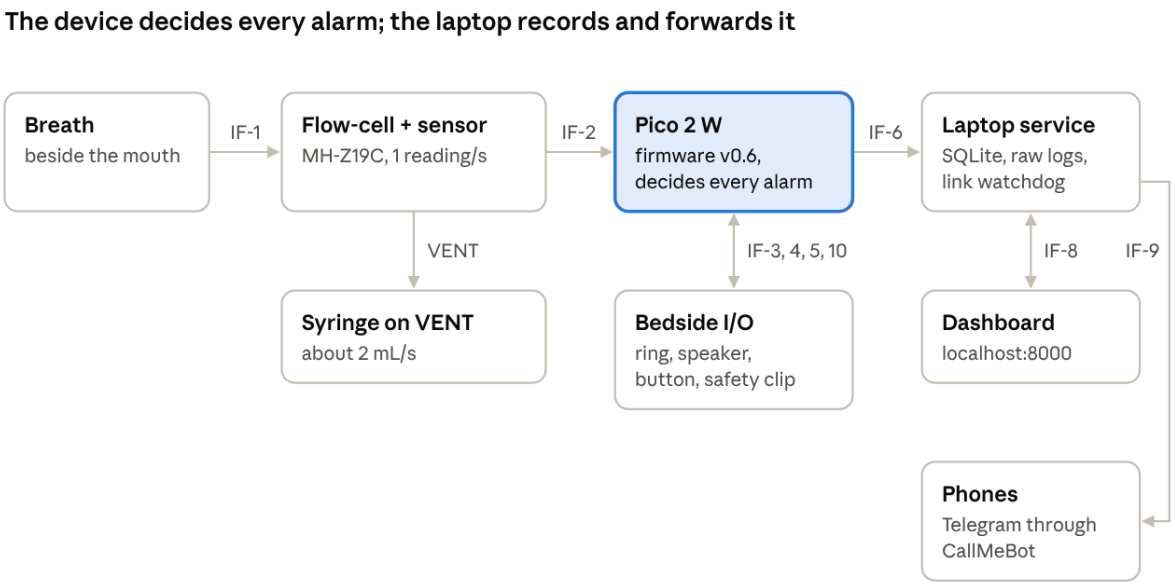


Figure 2. Block diagram. The interface identifiers match the interface table of D3.

The Pico is the only place a patient alarm is decided. The laptop service mirrors, records and forwards it, so a stopped service or a closed dashboard never silences the bedside alarm (test 7). The exception is power: the USB cable that carries IF-6 also powers the box (IF-11), so a pulled cable or a dead laptop turns it off (D8).

5.2 Alarm state chart

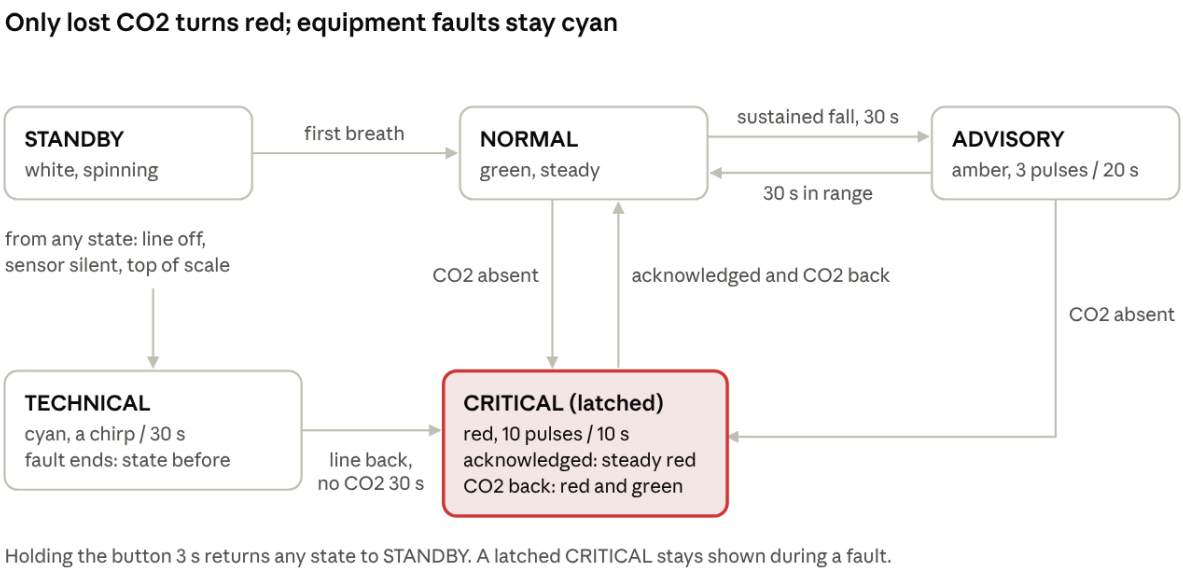


Figure 3. Alarm state chart of firmware v0.6 (`firmware/lib/alarm.py`).

A disconnected line goes to TECHNICAL, never CRITICAL (a CRITICAL already latched stays red); only after the line is back and 30 s pass with no CO2 does the device raise CRITICAL.

5.3 Wiring diagram

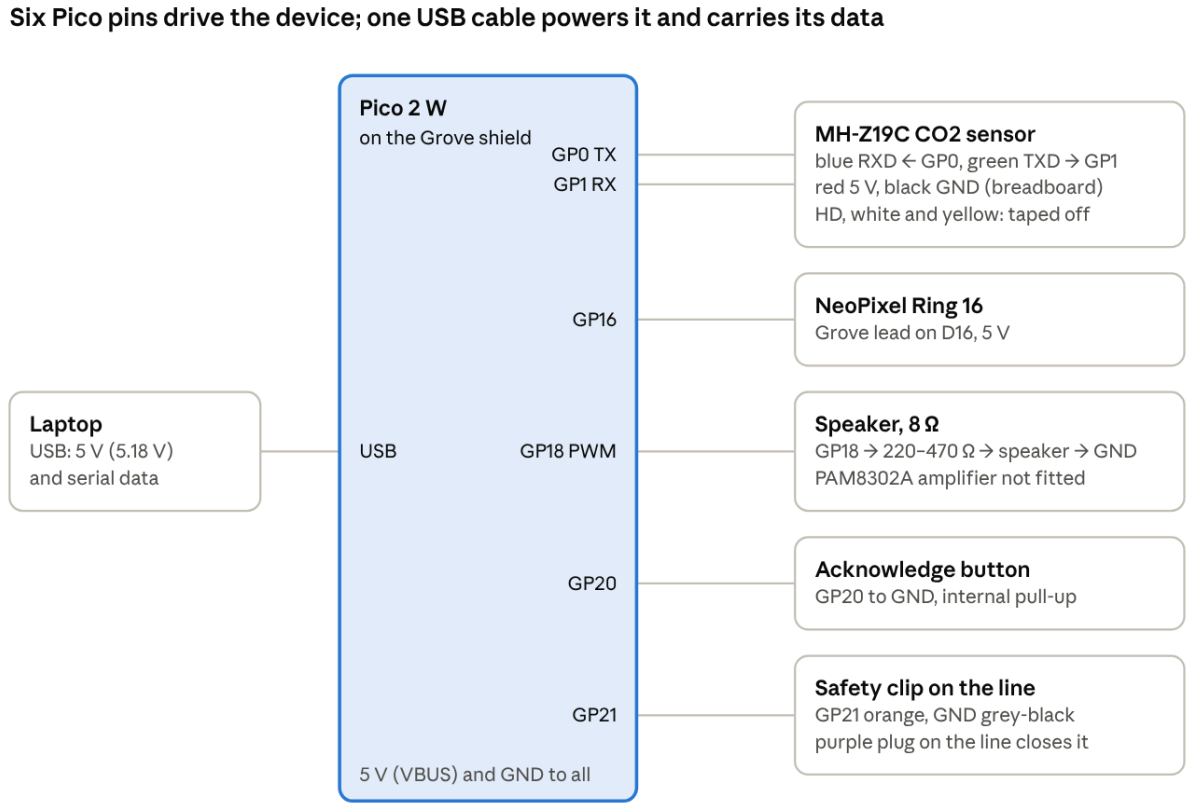


Figure 4. Wiring diagram, as built (`hardware/wiring.md`).

The speaker runs straight from GP18 through a resistor, so its tone is quiet. The kit's PAM8302A amplifier, once soldered, takes the same pin and makes it loud with no change to the firmware (D8).

5.4 Sequence: from a reading to a phone message

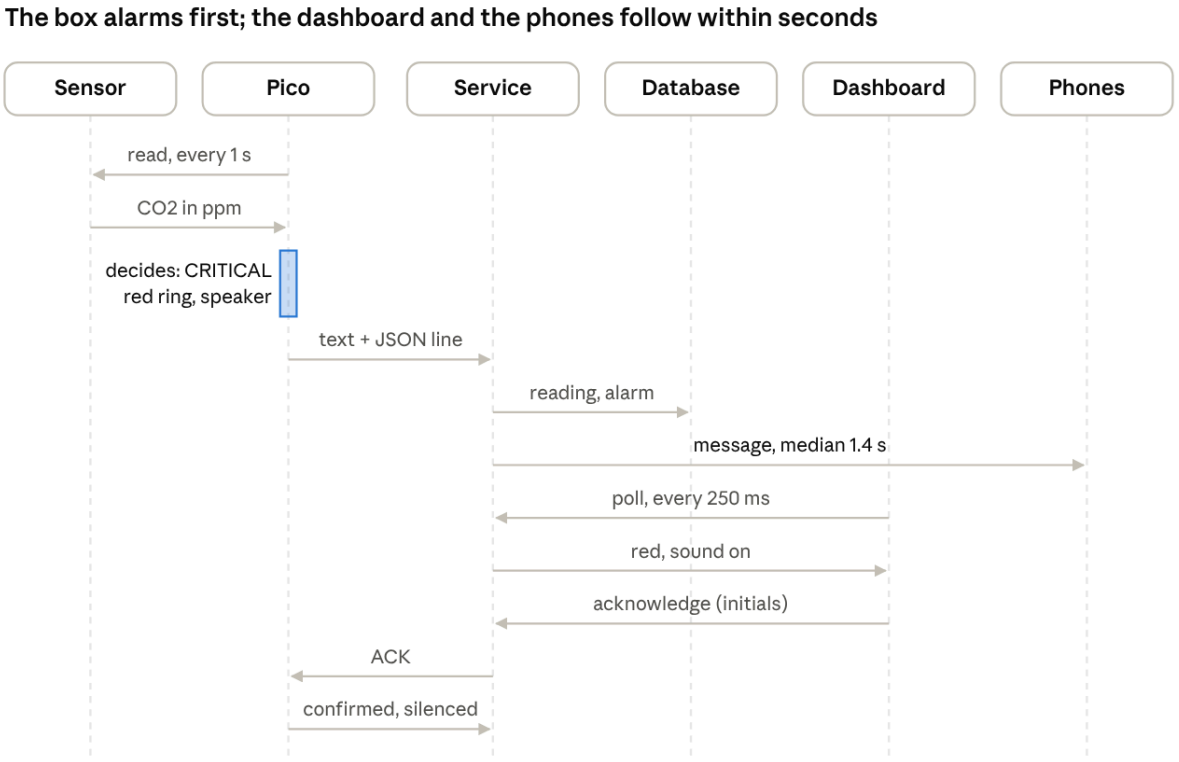


Figure 5. From a reading to a phone message. Message times from the notifications table, 1 to 4 Nov 2026.

The box has already turned red and sounded before the laptop sees the line. The service stores the alarm and sends the messages at once: across the 116 critical messages of 1 to 4 Nov, CallMeBot accepted each a median 1.4 s after the alarm reached the laptop (slowest 3.2 s). The dashboard shows the new state at its next poll, and an acknowledgement made there is confirmed by the box in its next reading.

SECTION 6, DELIVERABLE D5

Working system and documentation

The whole chain runs on the bench, from the sensor to a phone message, and everything that runs is in the repository with a README for each part. GitHub runs the 134 automated tests on every pull request; on a laptop without pyserial, 133 pass and the one serial-port test is skipped.

6.1 What is built

PART	STATE ON 4 NOV	WHERE
Firmware v0.6	CircuitPython 9.2.9, on the board since 1 Nov and used for every scored test: CR-1 presence and absence, STANDBY and the four alarm states, breathing rate and advisory, safety clip, top-of-scale rule, ring, speaker and button; one text line and one JSON line per reading	`firmware/`
Laptop service	Python 3.11 and Flask: serial ingest with a raw log of every line, rules engine with the 5 s link watchdog, SQLite store, notifier, REST layer	`service/`
Database	One SQLite file in write-ahead-log mode with the seven tables of D2. It holds participants' contact details, so it stays on the laptop and git ignores it	`data/co2monitor.sqlite`, laptop only
Dashboard	The box's state live (ring, alarm band, sounds), CO2, exhaled-CO2 state, breathing rate, line state, 10-minute CO2 and rate charts, alarm log, acknowledge with initials; refreshes 4 times a second	`dashboard/index.html`
Notification path	Opt-in sign-up page that stores the consent time; Telegram through CallMeBot; critical at once, others after 60 s, backups at 2 min, retries, at most 10 messages per person in 10 min	`service/notifier.py`, `dashboard/register.html`
Printed parts	Case base and lid, flow-cell tray and lid, drop-in cell clamp, light cap, barb-fit coupon; PLA, no supports	`hardware/cad/`
Bench tools	Start script, live view, scripted breath runs with spoken cues scored for ER-1 and ER-2, paced rate runs scored for ER-7 and ER-8, offline peak counting, calibration helper	`tools/`
Tests	134: firmware logic 85 (several replay the 22 and 25 Oct bench readings), service and REST layer 23, notifier 23, run scoring 3	`tests/`, `.github/workflows/tests.yml`
Records	The raw log of every bench session, results tables, calibration, lab notebook, photos, dashboard screenshots	`data/`, `docs/`

6.2 Firmware

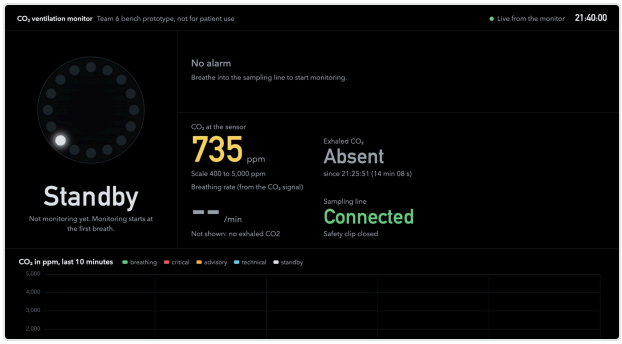
`code.py` and eight modules in `lib/`, about 1,700 lines of CircuitPython. The main loop reads, decides and reports once per second and detects nothing itself: each decision sits in a module that the laptop tests run without the board. An excerpt of the loop, from `firmware/code.py`:

```
while True:
    ppm = read_co2(orientation, idle)
    ...
    t = time.monotonic()
    changed = monitor.update(t, ppm)
    short, held = sound.take_presses()
    line.sample(t, line_pin.value)
    remote = sum(1 for c in commands() if c == "ACK")   # the dashboard's acknowledge button
    ...
    rate_banner = breathing.update(t, ppm, monitor.state, keep_alive)
    top_run = top_run + 1 if ppm is not None and ppm >= TOP_PPM else 0
    ...
    for action in alarm.update(t, monitor.state, monitor.reason, ppm, short + remote, held,
                               breathing.advisory, line.connected, top_run >= SATURATED_S):
        ...
    show.show(alarm.display())
    ...
    print(json.dumps({"v": VERSION, "t": int(t), "ppm": ppm, "cr1": monitor.state, ...}))
```

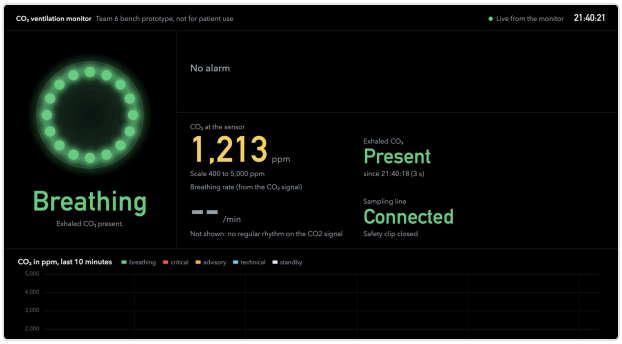
`monitor` is CR-1 (`lib/cr1.py`), `breathing` is CR-4 (`lib/cr4.py`), `line` is the safety clip and `alarm` the alarm manager. While the loop waits for the sensor or the next second, `idle()` keeps the ring flashing, the sound playing and the button and clip read, so a silent sensor cannot freeze the alarm, a bug found on 25 Oct and fixed with a test (D8).

6.3 Service, database and dashboard

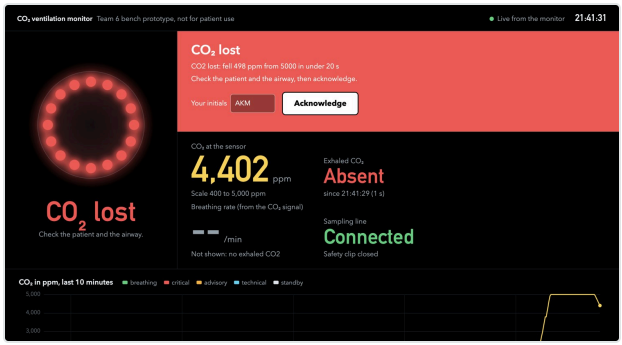
`bash tools/dashboard.sh` creates the Python environment on its first run, starts the service and opens the dashboard at localhost:8000. The service finds the Pico's serial port, again after a replug, saves every line to a dated raw log in `data/raw/`, parses it, mirrors the box's alarm into SQLite (D2) and serves the dashboard. The dashboard asks for the latest state 4 times a second and plays the same alarm patterns as the box. An acknowledgement typed there with the user's initials goes to the box as `ACK`, and the dashboard shows it as confirmed only when the box reports it.



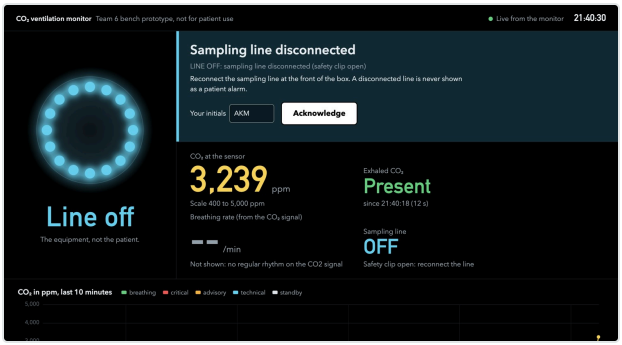
STANDBY



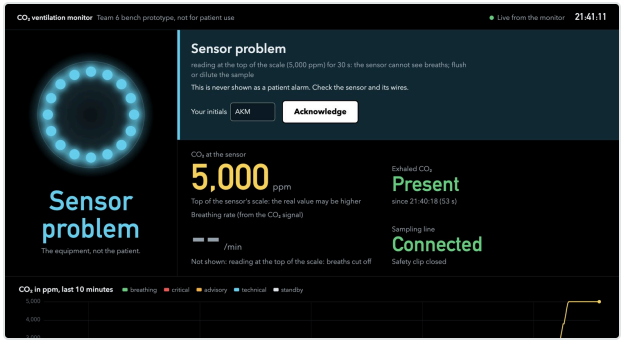
NORMAL



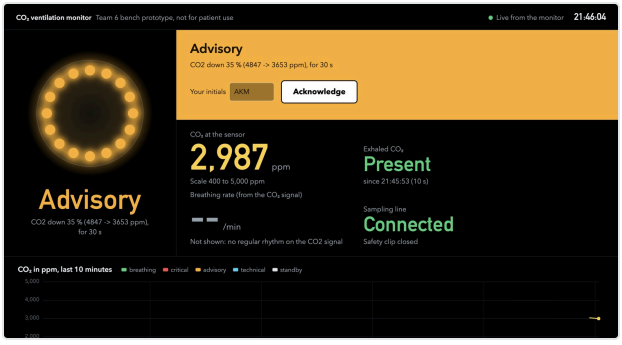
CRITICAL: CO2 LOST



TECHNICAL: LINE OFF



TECHNICAL: SENSOR AT ITS CEILING



ADVISORY: THE 3 NOV RECORDING

Figure 6. The dashboard in six states: STANDBY, breathing (NORMAL), CO2 lost (CRITICAL), line off and sensor at the top of its scale (TECHNICAL), and the dashboard view of the advisory recorded on 3 Nov, 18:36 (CO2 above room air down 35 %, 4,847 → 3,653 ppm, for 30 s). The screenshots were taken on 4 Nov to show the display; they were not captured during the recorded test runs. All eleven are in `docs/screenshots/`.

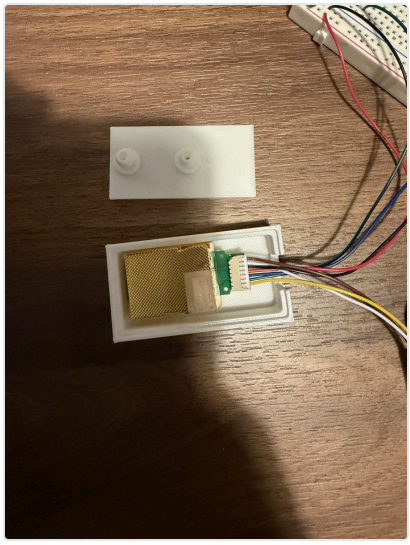
6.4 Notification path

A course participant opts in on the sign-up page with a name, role, Telegram user name, first list or backup, and a consent box; the service stores the consent time and sends a test message. Alarms then go through CallMeBot's Telegram API (IF-9) on the timing of D2. Every message is a row in `notifications` with its attempts and status, and a failed send shows on the dashboard. A message names the alarm and its reason, never a patient. From 1 to 4 Nov, CallMeBot accepted 116 critical messages a median 1.4 s after the alarm reached the laptop (slowest 3.2 s); their arrival on the phones was not timed, and one message was held back by the 10-messages-in-10-minutes limit (D7, D8).

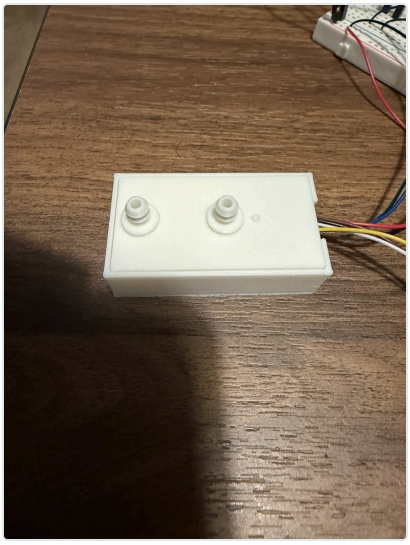
6.5 Printed parts

Every part prints in PLA without supports. The case was sized from caliper measurements of the kit parts, the flow-cell (version 1) from the sensor's datasheet. The flow-cell stands on edge against the inside of the front wall, so the line pushes onto SAMPLE from outside: no tubing inside the box, and lines of other lengths swap without opening it.

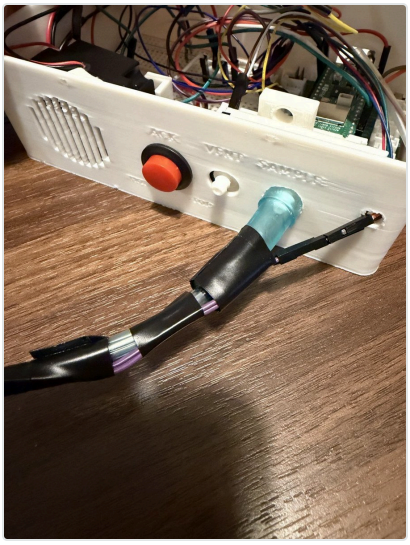
PART	SIZE (MM)	WHAT IT DOES
Case base	152 × 94 × 50	Front panel: speaker grille, ACK button, VENT and SAMPLE, engraved labels. Inside: a pocket for the breadboard, four posts for the shield, USB out of the back
Case lid	152 × 94 × 6.6	Snaps on; a 41 mm window and collar hold the light ring under the light cap
Flow-cell tray and lid	51.3 × 27.5 × 15.6 (tray)	The sealed cell around the sensor; the inlet barb sits over the sensor's gas window
Cell clamp	49 × 41 × 8.5	Drops in behind the cell and keeps its barbs in the front-panel holes
Light cap	44 × 44 × 15.3	Diffuser over the ring
Barb-fit coupon	60 × 14 × 9	Five test barbs, 4.8 to 6.4 mm, printed first to choose the barb size; the tubing held firmly on the lid's 5.6 mm barbs



FLOW-CELL OPEN, SENSOR IN ITS TRAY



FLOW-CELL CLOSED AND TAPED



FRONT PANEL: LINE ON SAMPLE, SAFETY CLIP

Figure 7. The printed flow-cell open with the sensor in its tray, and closed (25 Oct); the case front panel with the line on SAMPLE and the safety clip (1 Nov).

6.6 Deployment

From a clean checkout, the README's getting-started takes five steps.

1. Wire the board as in `hardware/wiring.md` (D4). The sensor needs 5 V, and its brown calibration wire stays insulated.
2. With CircuitPython 9.2.9 on the Pico 2 W, copy `firmware/lib/` onto the CIRCUITPY drive first, then `boot.py` and `code.py`. The board restarts, shows every alarm colour (lamp test) and chirps.
3. On the laptop, from the repository folder, run `bash tools/dashboard.sh`. The first run installs Flask, pyserial and certifi into `.venv/`, and the browser opens http://localhost:8000; click "Turn on alarm sound" once.

4. To get the alarms on a phone, sign up at http://localhost:8000/register and press Start on CallMeBot's Telegram bot once.
5. Push the line onto SAMPLE with the safety-clip plug on it and the syringe onto VENT, then do the finger test: with the tube end closed, the syringe must be hard to pull. The ring circles white (STANDBY) through the 75 s warm-up and until the first breath turns it green.

The README's troubleshooting table lists the eleven messages a new user is most likely to see, each with what it means and what to do. Without the board, `bash tools/dashboard.sh --replay data/raw/co2_2026-10-25_011421.log --speed 5` plays a recorded session through the same service, and `python -m pytest tests/` runs the tests.

6.7 Live demonstration (about 5 minutes)

1. Power-up: the lamp test shows every alarm colour, the speaker chirps, and the ring circles white while the sensor warms up (STANDBY).
2. Breathe at the tube end, 10 cm from the mouth, while pulling the syringe steadily: green within 3 breaths, and the CO2 trace rises on the dashboard.
3. Stop breathing and keep pulling: red within about 15 s (the median of the scored runs), 10 pulses from the box and the dashboard, and a Telegram message on every registered phone.
4. Acknowledge on the dashboard with initials: the box confirms, its sound stops for 2 min, and red clears to green once breathing starts again.
5. Pull the line off SAMPLE: cyan "Line off", never red. Push it back and breathe: green.
6. Stop the laptop service (Ctrl+C) during a red alarm: the box keeps flashing and sounding, and the dashboard says "No service".

SECTION 7, DELIVERABLE D6

Verification plan

Each ER has a bench test whose method, apparatus, sample size and pass criterion were fixed before its scored runs. Every reading of every run is saved, and the CO2 runs are scored by script from the saved log, not by eye. Abdul Aziz Mourad led the testing, with Michele Frederike and Szirina Ismail helping to run several tests, including tests 4 and 5; he signed off each result against its saved log.

Apparatus (all tests). The device on firmware v0.6 with its sealed flow-cell; the 200 mL syringe on VENT at about 2 mL/s; the 0.5 m vinyl line (test 6 also 1 m of vinyl and the kit's 2.1 m oxygen tubing); the laptop service recording every reading; the spoken-cue scripts in `tools/`; three registered phones for test 7. The breath source is a team member at the open end of the line, without a mask. Unless a row says otherwise, the tube end is 10 cm from the mouth and the syringe is pulled at about 2 mL/s.

ER	TEST	METHOD	SAMPLE	PASS IF	SIGNED OFF BY
ER-1	5	`scripted_run.sh --rate 15 --breaths 10 --rest 60`: a spoken cue per breath; green timed from the first cue	10 runs	Green by the third breath in 10 of 10	A. A. Mourad
ER-2	5, 7	Test 5 runs: red timed from the stop cue. Test 7: every critical alarm of the bench sessions, from the alarm and notification tables; the service stopped during an alarm	10 runs; every critical alarm	Red within 30 s in 10 of 10; dashboard within 2 s; every phone messaged within 10 s	A. A. Mourad
ER-3	6	`scripted_run.sh --breaths 1`: one 2 s breath into the tube end at the lips on the cue; first rise (30 ppm above the 10 s before the cue), green, red and 10–90 % rise read from the log	3 trials per line length (0.5, 1 and 2.1 m), one person pulling	Mean and range published per length, with the breath-by-breath decision	A. A. Mourad
ER-4	1	Outdoors with the 2.1 m kit tubing, the tube end upwind and 2 m from any face, 40 mL drawn every 2 min, the analysis window fixed before looking. Then the reference capnograph and the device on the same gas at 4 to 5 levels	10 min of readings; 4 to 5 levels	Drift within ±50 ppm in 10 min; each level within ±(50 ppm + 5 %) after the one-point zero	A. A. Mourad
ER-5	2, 3	Test 2: the line pulled off SAMPLE and replaced ten times. Test 3: the line pulled off for over a minute while the cell's breath drains, then replaced with no breath	10 pulls; 1 drain run	Line off within 2 s every time; 0 CRITICAL while off	A. A. Mourad
ER-6	4	The 2.1 m kit tubing pinched shut by hand for 60 s during monitoring, the draw continuing	1 trial, sealed cell	TECHNICAL within 30 s and 0 CRITICAL	A. A. Mourad
ER-7	rate runs	`rate_run.sh`: paced spoken cues per phase, tube end 15 cm from the mouth, about 1 mL/s, scored by `score_rate_run.py`	5 runs at each of 6, 10, 15 and 20/min	In every phase from 50 s in: a rate shown at least 80 % of the time, its median within tolerance, none below 80 % of the paced rate	A. A. Mourad
ER-8	rate runs	Paced runs slowing from 15 to 6/min; 10 min of steady paced breathing	10 slowing runs; 5 steady runs	Advisory within 90 s in 10 of 10; none in the steady runs	A. A. Mourad

Protocol for the scored CO2 runs. Test 5 follows `data/runs/test5_protocol.md`, fixed before its first run; test 6 used the same steps with one breath per trial.

1. Finger test: with the tube end closed, the syringe must be hard to pull (the cell is sealed).
2. Flush the cell with room air until STANDBY shows a steady room level.
3. Start the script; it saves the log, with the time of every cue, in `data/raw/`.
4. Pull steadily at about 2 mL/s (test 5: one 10 mL mark every 5 s, from 10 s before the first cue), with no syringe swap during the run.
5. Breathe out on each cue: in test 5 the tube end is taped to a ruler 10 cm from the mouth; in test 6 one 2 s breath goes into the tube end at the lips. After the stop cue, close the mouth, turn away and keep pulling.

6. Score by script from the saved log: `scripted_run.sh` adds each run to `data/runs/results.csv`, and the test 6 timings are in `data/runs/tube_length_results.csv`.

The reference capnograph is used for calibration and validation only, never as a live channel during a logging run.

SECTION 8, DELIVERABLE D7

Verification report

Three ERs pass (ER-1, ER-3, ER-5), one is partly met (ER-4), three are not met (ER-2, ER-6, ER-7) and one is not verified on the bench (ER-8), each with its measured cause. The scored tests ran from 2 to 4 Nov 2026 on firmware v0.6. Every number here comes from the saved logs and tables in `data/`, except the phone timings, which come from the laptop's database.

8.1 Calibration and drift (ER-4)

No reference instrument was available before submission, so the calibration is a one-point zero against outdoor air; above the zero, the span is the manufacturer's [4].

QUANTITY	VALUE
Reference	Outdoor air: NOAA Mauna Loa monthly mean for September 2026, 426.07 ppm [8], taken as 426 ppm with 0 to +30 ppm allowed for city air
Recording	4 Nov, outdoors at night, 21:28:00 to 21:37:32 (window fixed before looking): 573 readings, no alarms
Device reading	Mean 720.5 ppm, median 722, SD 16.3, range 689–751
Zero offset	+294 ppm (+264 to +294 allowing for city air)
Correction	True CO2 = reading − 294 ppm, applied in the analysis. The sensor's zero command was not used, so the earlier logs stay comparable
Drift	Least-squares slope −2.25 ppm/min: −21.5 ppm over the 9 min 32 s record, −23 ppm extrapolated to 10 min (limit ±50 ppm)
Residuals about the drift line	SD 15.0 ppm, from −28 to +34 ppm
Span above the zero	±(50 ppm + 5 % of reading), from the manufacturer [4]; not checked

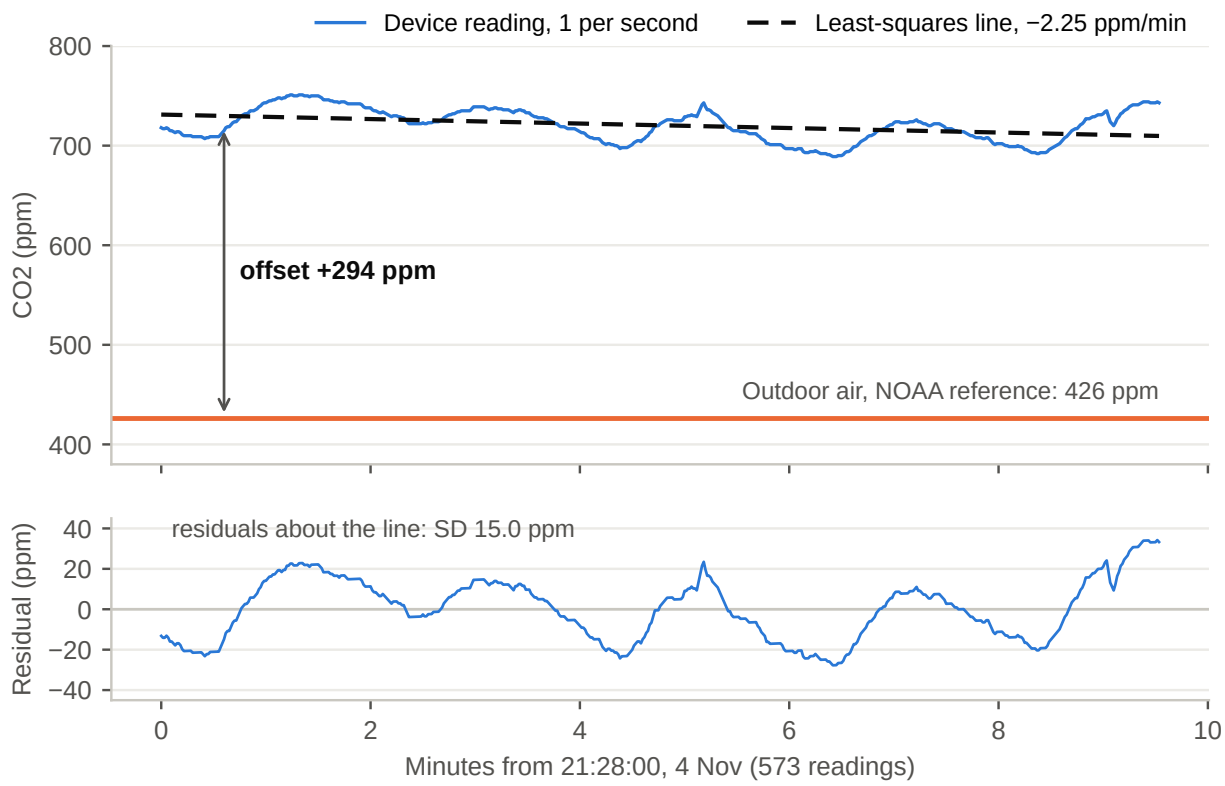


Figure 8. The outdoor zero recording of 4 Nov: 573 readings, the least-squares drift line and the residuals about it.

A first check by an open window on 2 Nov read 756 ppm; outdoors, with the cell sealed, the reading was 720 ppm, so the window set-up does not explain the offset; the sensor, the cold and the gas path cannot be separated without a reference gas. The calibration curve with residuals at 4 to 5 levels that ER-4 asks for needs a reference gas or the simulation-lab capnograph; neither was available, so ER-4's accuracy half is not verified. The alarms work on changes in the reading, so the offset changes no alarm result.

8.2 Alarm latency (ER-1, ER-2)

Test 5, 4 Nov: ten scripted runs of 10 breaths at 15/min through the 0.5 m line, the tube end 10 cm from the mouth.

RUN	GREEN AFTER THE FIRST CUE (S)	BREATH	RED AFTER THE STOP (S)	PEAK (PPM)	AT THE 5,000 PPM CEILING
1	6	2	24	5,000	yes
2	9	3	21	2,228	no
3	7	2	15	5,000	yes
4	7	2	15	3,598	no
5	8	3	33	5,000	yes
6	6	2	16	3,526	no
7	7	2	13	4,167	no
8	8	3	8	3,756	no
9	7	2	9	4,595	no
10	8	3	13	3,328	no

Green came at breath 2 or 3 in every run, 6–9 s after the first cue (mean 7.3 s). Red came 8–33 s after the stop (median 15 s, mean 16.7 s, SD 7.5): 8–21 s in the seven runs that stayed below the ceiling (mean 13.6 s), and 15, 24 and 33 s in the three that reached it. In run 5 the reading sat at 5,000 ppm from 1 s before the stop until 28 s after it, so the sensor could not see the fall; red came 5 s after the reading left the ceiling. The designed fixed dilution (D2) keeps the sample below it.

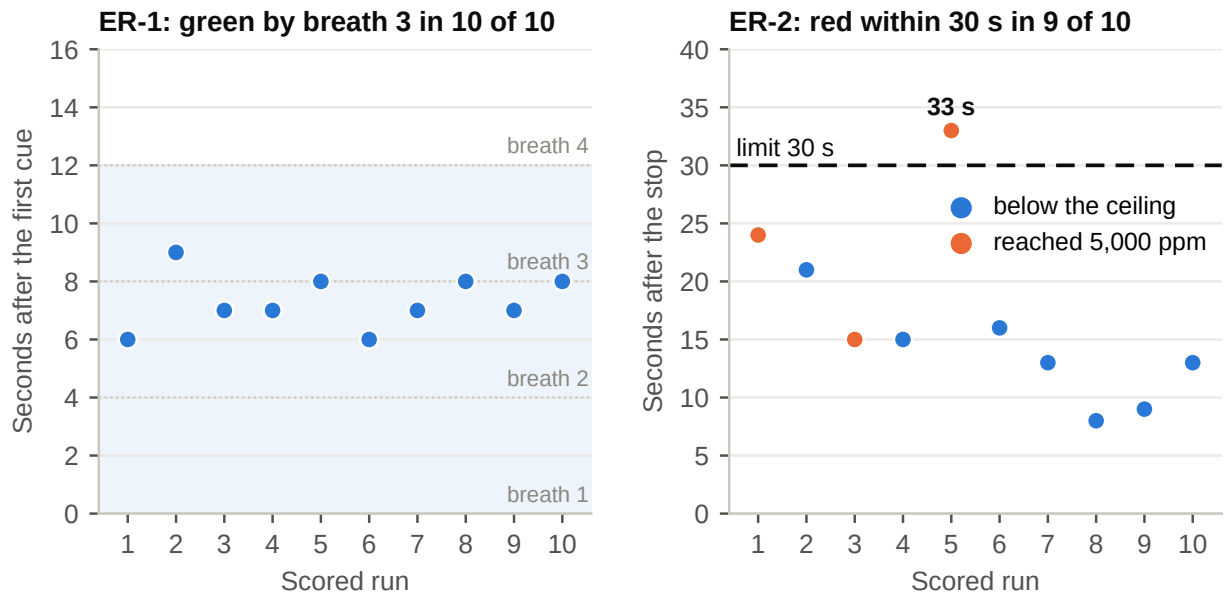


Figure 9. Test 5, run by run: green after the first cue (left) and red after the stop (right). Orange marks the three runs that reached the sensor's 5,000 ppm ceiling.

The laptop received every reading within the second it was made (in the ten test 5 logs, all 1,231 steps between readings were 1 s on both clocks), and the dashboard asks for the state every 250 ms, so the dashboard shows an alarm within 2 s by design; this was not timed alarm by alarm on screen. Phones: from 1 to 4 Nov, CallMeBot accepted 116 critical messages a median 1.4 s after the alarm reached the laptop, the slowest at 3.2 s; their arrival on the phones was not timed. One critical message (3 Nov, 17:08) was never sent: after a run of alarms during bench set-up, the limit of 10 messages per person in 10 minutes held it back (D8).

8.3 Response time against sampling-line length (test 6, ER-3)

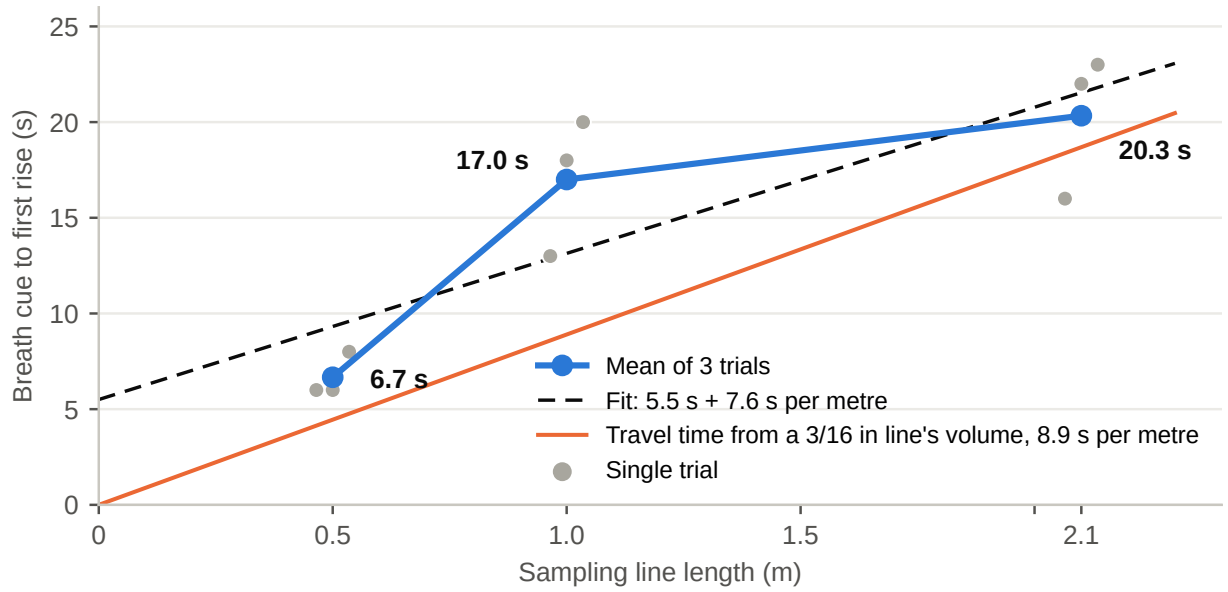


Figure 10. Breath cue to first rise against line length (test 6): single trials, the mean per length, the straight-line fit, and the travel time a 3/16 in line's volume predicts at 2 mL/s.

LINE	FIRST RISE (S)	GREEN (S)	RED AFTER THE BREATH (S)	10–90 % RISE (S)	RED AFTER THE PEAK (S)
0.5 m	6.7 (6–8)	7.3 (7–8)	39.7 (39–40)	14.3 (14–15)	10.3 (9–11)
1 m	17.0 (13–20)	18.0 (14–21)	50.3 (45–58)	13.0 (11–15)	11.7 (10–14)
2.1 m	20.3 (16–23)	23.3 (22–24)	56.7 (53–62)	14.3 (12–17)	10.3 (10–11)

Mean (range) of 3 single-breath trials per length, one person pulling, sealed cell (2 to 3 Nov). Times are from the breath cue, so they include the reaction to the cue. The line adds delay, not slowness: the 10–90 % rise is 13.9 s (SD 1.8) at every length, and once the peak has passed, the alarm logic reacts in 9–14 s.

A straight-line fit gives a first rise of 5.5 s plus 7.6 s per metre of line; a 3/16 in line holds 17.8 mL per metre, which at 2 mL/s would add 8.9 s per metre; the 2.1 m line is the mask kit's oxygen tubing, whose bore was not measured, so that prediction holds strictly for the vinyl lines. The 1 m set ran last, after the 2.1 m set, and its rises were the smallest (mean 1,128 ppm, against 2,208 and 1,930 ppm), so a slower hand pull at the end of the session is the likely cause of its long 17.0 s (D8). Breath-by-breath capnography would need a 10–90 % rise shorter than one breath, 4 s at 15/min; at 13.9 s this device is a breath-averaged trend monitor, and ER-1 to ER-8 are written for one.

8.4 Labelled fault and event dataset

Every fault injected on the bench and every alarm event used as evidence, labelled from the logs. The last five rows were seen outside the protocol, not staged; they are the evidence for D8.

LABEL	WHAT WAS DONE OR SEEN	N	EXPECTED	DEVICE RESPONSE
Line disconnected	Line pulled off SAMPLE from STANDBY and replaced, off 8–20 s each (test 2, 2 Nov)	10	TECHNICAL within 2 s	TECHNICAL within the 1 s reading, 10 of 10; 0 CRITICAL

LABEL	WHAT WAS DONE OR SEEN	N	EXPECTED	DEVICE RESPONSE
Line off while CO2 drains	Line off 74 s while the reading fell from 1,190 to 847 ppm (test 3)	1	TECHNICAL, never CRITICAL	TECHNICAL throughout; 0 CRITICAL
Line back, no breath	Line replaced with no breath (test 3)	1	CRITICAL 30 s later	CRITICAL at 31 s
Line blocked	Line pinched for 60 s during monitoring, sealed cell (test 4, 3 Nov)	1	TECHNICAL within 30 s	CRITICAL 35 s after the pinch; no TECHNICAL
Breathing stopped	End of 10 breaths at 15/min (test 5, 4 Nov)	10	CRITICAL within 30 s	CRITICAL 8–33 s after the stop; 9 of 10 within 30 s
Steady breathing	10 breaths at 15/min (test 5)	10	No alarm	0 CRITICAL, 0 ADVISORY
Sensor at its ceiling	Reading at 5,000 ppm for 30 s (test 5 runs 1 and 5; the paced rate run, 4 Nov)	3	TECHNICAL	TECHNICAL, 3 of 3
Sensor silent	Sensor data wire pulled during monitoring (25 Oct, firmware v0.4)	1	TECHNICAL	TECHNICAL within 1–2 s
Laptop service stopped	Ctrl+C during a CRITICAL (test 7, 2 Nov)	1	Box keeps alarming	Box kept flashing and sounding; dashboard "No service"
Box restarted by itself over USB	2 Nov, at 16:42, 18:32 and about 23:40	3	Dashboard "No data" TECHNICAL after 5 s	Back in STANDBY with its warm-up each time; nothing was latched then, but a latched alarm would have been lost
Slow fall with no draw	One heavy breath with the syringe not pulled (3 Nov, 18:36)	1	CRITICAL within 30 s of the stop	ADVISORY ("CO2 down 35 %") for at least 4 min 23 s, then CRITICAL
Room CO2 or nearby breath at the line end	STANDBY, nobody breathing into the line (2 Nov, 17:11 and 18:34)	2	No alarm	False CO2 present (green); at 18:35 a false CRITICAL followed
Draw pulled in steps	35 mL pulled in one go every 10 s (2 Nov, 17:05 to 17:15)	3	No alarm	False CRITICAL
Breathing with no draw	Breathing through the 0.5 m line with no syringe on VENT (3 Nov, 00:48 to 00:50)	2	No alarm	False CRITICAL

8.5 Sensitivity and specificity

Scored on the ten test 5 runs, each with one breathing period and one stop, and on the injected equipment faults. Ten runs give wide intervals (Wilson 95 %).

MEASURE	RESULT
Sensitivity: stops that raised CRITICAL within 30 s	9 of 10, 0.90 (0.60–0.98); 10 of 10 within 33 s
Specificity: breathing periods with no false CRITICAL	10 of 10, 1.00 (0.72–1.00); an advisory cannot arm in 40 s of breathing
CO2 presence shown by the third breath	10 of 10, 1.00 (0.72–1.00)
Equipment faults while monitoring (v0.6), shown as TECHNICAL, never as apnoea	4 of 5, 0.80 (0.38–0.96): line off during a fall 1 of 1, sensor at its ceiling 3 of 3, blocked line 0 of 1. Also detected: line off from STANDBY, 10 of 10; sensor silent, 1 of 1 (firmware v0.4)
False CRITICAL in the 2 min paced run	0
Falsely low breathing rate shown, in all runs and replays	0

8.6 Breathing rate and advisory (ER-7, ER-8)

The rate is resolved only when the breaths reach the sensor directly. Replayed through firmware v0.6, the 22 Oct paced run (bare sensor, 15/min, no line) shows 14.9–15.7/min from 31 s into the run, and offline peak counting gives 15.0/min (15 bumps). Through the line and the sealed cell the rate is "not resolved". In all ten test 5 runs the checks rejected every candidate rhythm, withholding candidates of about 4–8/min against a true 15/min. In the paced bench run of 4 Nov (10/min for 120 s) the reading reached the 5,000 ppm ceiling 37 s after the first cue and stayed there, so the box showed "breaths cut off". No run or replay showed a falsely low rate; on the replay the rate stayed near 15/min for 10 s after breathing stopped. The planned paced runs at 6, 10, 15 and 20/min were not completed: through this line and sensor, the ripple that carries the rate is too small, or cut off at the ceiling (D8).

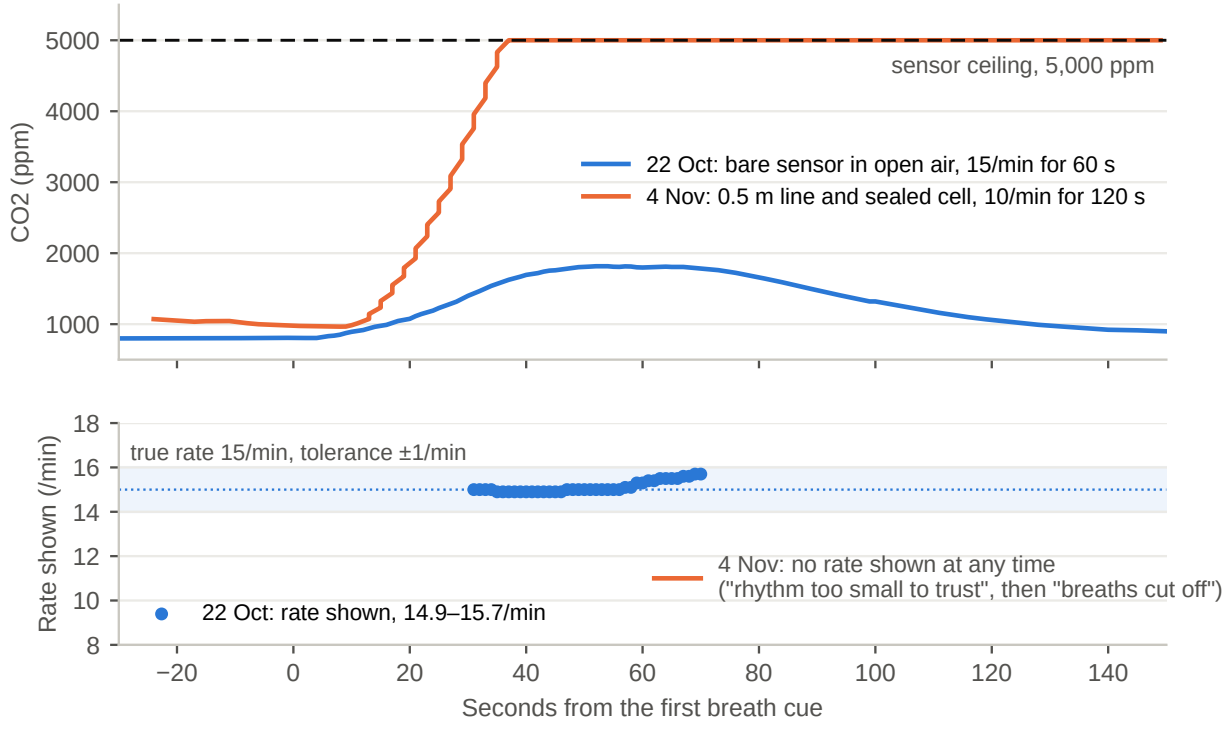


Figure 11. Breathing rate. Top: the CO2 reading of the 22 Oct paced run (bare sensor, 15/min for 60 s) and of the 4 Nov paced run (0.5 m line and sealed cell, 10/min for 120 s). Bottom: the rate the device showed.

One ADVISORY was recorded on the bench, on 3 Nov at 18:36. After one heavy breath with the syringe not being pulled, the reading fell slowly and the box showed amber: "CO2 down 35 % (4847 → 3653 ppm), for 30 s", a fall of 35 % in the CO2 above room air. The recording began after the amber had started, so the time from the fall to the advisory is not known. The amber held for at least 4 min 23 s, until the reading had fallen back near its starting level and the box went red: with no draw, a stop shows first as an advisory (D8). In a simulation sweep of 26–27 Oct (recorded with ER-8 in `docs/D1-engineering-requirements.md`), a rate falling from 15 to 6/min gave the advisory 60–78 s after the slowing (limit 90 s), CO2 halving over 2 min gave it 45–52 s after the fall reached 30 % (limit 60 s), and 135 ten-minute runs of steady breathing gave no false advisory. No bench run tested the advisory's specificity: it needs about 2 min of steady breathing to learn a baseline, longer than the breathing in test 5, and the paced run reached the sensor's ceiling after 37 s. The paced slowing runs were not done.

8.7 Result against every requirement

ER	RESULT	VERDICT
ER-1	Green by the third breath in 10 of 10 runs, 6–9 s after the first cue (mean 7.3 s)	Pass
ER-2	Red within 30 s of the stop in 9 of 10 runs (8–33 s, median 15 s); the late alarm, 33 s, came while the sensor sat at its 5,000 ppm ceiling. Dashboard within 2 s by design. Phones: CallMeBot accepted 116 critical messages (median 1.4 s, slowest 3.2 s), arrival not timed; one held back by the 10-in-10-minutes limit	Not met (9 of 10)
ER-3	First rise 6.7 s (0.5 m), 17.0 s (1 m) and 20.3 s (2.1 m); 10–90 % rise 13.9 s at every length, so breath-by-breath capnography is not claimed	Pass
ER-4	One reading per second across 400–5,000 ppm; drift −21.5 ppm over the 9 min 32 s record (limit ±50 ppm in 10 min); zero offset +294 ppm, corrected in the analysis; accuracy at 4–5 reference levels not measured, for want of a reference instrument	Partly met
ER-5	Line off within the 1 s reading in 10 of 10 pulls from STANDBY, 0 CRITICAL; 74 s off while the CO2 fell, 0 CRITICAL	Pass
ER-6	The pinched line raised CRITICAL 35 s after the pinch, never TECHNICAL	Not met
ER-7	14.9–15.7/min against a true 15/min on the bare-sensor replay (15.0/min by peak counting); "not resolved" through the line and cell in every bench run; never a falsely low rate; the rate, or why it is withheld, is stored with every reading	Not met
ER-8	Recorded once on the bench, onset not captured; in simulation within the limits, with no false advisory in 135 steady runs; the paced bench runs not done	Not verified on the bench

SECTION 9, DELIVERABLE D8

Failure modes and mitigation

Twenty failure modes were considered, eighteen of them found on the bench. Fifteen have a mitigation implemented and verified; five have a designed fix only. The six still rated 6 after mitigation trace to four causes: no pump and no line-pressure sensing (FM-2, FM-7), a sensor too slow and too low in range for breath (FM-5, FM-6), the box's dependence on USB power (FM-11), and the service's message limit (FM-16), whose designed fix is a small change not made before submission.

Rating. Severity (S): 3 = a missed, late or silenced patient alarm; 2 = a false or delayed alarm, or one alarm channel lost; 1 = a wrong level with no alarm effect. Likelihood (L): 1 = seen once or never, and unlikely in use; 2 = seen more than once, or likely in use; 3 = seen in most sessions, or expected in routine use. Risk = S × L, before and after mitigation. Any risk of 6 or more needed a mitigation implemented and verified, or a designed fix with its cause.

FM	AREA	FAILURE	EFFECT ON THE PATIENT ALARM	S × L BEFORE	MITIGATION	VERIFIED	S × L AFTER
FM-1	Use	Sampling line disconnected	Falling CO2 would read as apnoea	3 × 3 = 9	Safety clip on the line: TECHNICAL "Line off", never CRITICAL. The first clip, inside the case, did not open when the tube came off (1 Nov); rebuilt outside, so the tube pulls the plug along its length	Tests 2 and 3: 11 of 11 shown as Line off, 0 CRITICAL	3 × 1 = 3
FM-2	Use, environment	Sampling line blocked (kink, water, pinch)	Shown as apnoea instead of a fault	3 × 2 = 6	None on the device: the kit has no means to sense line pressure or flow. Designed: pump-current sensing, or a pressure sensor between cell and pump, giving cyan "Line blocked"	Test 4: CRITICAL 35 s after the pinch, no TECHNICAL	3 × 2 = 6
FM-3	Hardware	Air leak into the gas path at the cell's wire exit	Diluted, smaller rises: presence late or missed	3 × 2 = 6	Leak found on 2 Nov at 22:16 and sealed; finger test before every run; the gas-path tests repeated with the sealed cell	One breath raised the reading by 201 ppm with the leak open, by 2,018–2,463 ppm sealed (0.5 m line)	3 × 1 = 3
FM-4	Use, environment	CO2 from people or room air at the open line end	False "CO2 present"	2 × 2 = 4	Line end at the patient's mouth only; no talking near the line during a run; flush to a steady baseline first	Two false "present" in STANDBY during set-up (2 Nov); none before the first breath in the 10 scored runs	2 × 1 = 2
FM-5	Hardware (sensor range)	Undiluted breath above the 5,000 ppm ceiling	Breaths invisible; a stop is seen late	3 × 3 = 9	TECHNICAL after 30 s at the top of the scale; a fall from the ceiling goes CRITICAL as soon as it is called; tube end held 10 cm from the mouth. Designed: fixed room-air dilution; a 10,000 ppm module	TECHNICAL 3 of 3 times; 3 of the 10 scored runs still reached the ceiling, one with red at 33 s	3 × 2 = 6
FM-6	Hardware (sensor speed)	10–90 % rise of 13.9 s, longer than one breath	No breath-by-breath signal; no rate through the line	2 × 3 = 6	The rate shows "not resolved" rather than a guess; the ERs are written for a trend monitor. Designed: a faster NDIR module reading to 100,000–200,000 ppm (10–20 %)	"Not resolved" in every bench run; no falsely low rate in any run or replay	2 × 3 = 6
FM-7	Use	Gaps in the sample draw, or no draw (hand syringe)	The reading sags: false CRITICAL. With no draw a stop falls slowly: ADVISORY first, CRITICAL minutes later	3 × 3 = 9	Steady pull, one 10 mL mark every 5 s, no syringe swap during a run. Designed: a pump at a fixed flow	No false alarm in the 10 scored breathing periods. Outside the protocol: 3 false CRITICAL with the draw in steps, 2 with no draw, and an advisory for at least 4 min 23 s before red with no draw	3 × 2 = 6

FM	AREA	FAILURE	EFFECT ON THE PATIENT ALARM	S × L BEFORE	MITIGATION	VERIFIED	S × L AFTER
FM-8	Software	The absence step (50 ppm, or 10 % of the rise) is small against a weak, drifting signal	False CRITICAL while breathing	2 × 2 = 4	None in firmware v0.6. Designed: a step relative to the breath, such as 30 % of the last rise, replayed on the saved logs before it is loaded	Two false CRITICAL with no draw (3 Nov)	2 × 2 = 4
FM-9	Use	Hand-drawn flow varies by operator	The delay varies with the person pulling	2 × 3 = 6	One person pulls for a whole test series, counting the syringe marks. Designed: a pump at a fixed flow	Same 1 m line: first rise 7–8 s with a second person pulling, against 13–20 s in the test 6 set, which ran last and had the smallest rises	2 × 2 = 4
FM-10	Software	Laptop service or dashboard stops	No dashboard, no phone messages	3 × 2 = 6	The box decides and sounds every alarm itself; the dashboard shows "No service"	Test 7: the box kept flashing and sounding with the service stopped	2 × 2 = 4
FM-11	Hardware	USB link drops, or the cable is pulled	Data gap; with the cable out the box is off and silent	3 × 2 = 6	Link watchdog: TECHNICAL "No data" after 5 s, phones after 60 s; the service finds the port again after a replug. Designed: a power source of the box's own, and the latch kept across a restart	The box restarted three times on 2 Nov, each time in STANDBY, and came back through its warm-up; a latched alarm would have been lost	3 × 2 = 6
FM-12	Software	Ring and sound freeze while the sensor is silent	No light or sound while a patient alarm stands	3 × 2 = 6	The driver keeps the ring, sound and button running while it waits for the sensor; a laptop test reproduces the bug	Found on 25 Oct with a data wire pulled; after the fix the ring kept running, and the test passes	3 × 1 = 3
FM-13	Hardware	Speaker wire comes loose	Silent alarm, not detected by the box	3 × 2 = 6	Start-up chirp to listen for at every power-up; jumper re-seated and taped. Designed: soldered wiring	Found on 1 Nov after three unheard alarms; fixed the same evening	3 × 1 = 3
FM-14	Hardware	Speaker quiet, amplifier not fitted	Alarm missed in a noisy room	2 × 2 = 4	Dashboard sound and phone messages in parallel. Designed: solder the kit's PAM8302A	Test 7: the box faint, the dashboard and all three phones alerted	2 × 1 = 2
FM-15	Software, network	Message not delivered (provider down, bot not started)	A phone is not told	2 × 2 = 4	Three tries (again after 10 s and 30 s); failures shown on the dashboard; the local alarm unaffected	Teammates' first messages arrived only once each pressed Start (1 Nov); automated tests	2 × 1 = 2
FM-16	Software	The 10-in-10-minutes limit holds back a critical message	A phone is not told of a real alarm	3 × 2 = 6	None in the service yet. Designed: never limit CRITICAL; limit only repeats of the same alarm	3 Nov, 17:08: one critical message held back	3 × 2 = 6
FM-17	Hardware	Sensor stops answering, sticks at 400 ppm, or jumps faster than CO2 can	Readings meaningless	3 × 1 = 3	FAULT and TECHNICAL, a 3 min re-warm, then 30 s for a breath; a latched CRITICAL stays red	Data wire pulled (25 Oct): TECHNICAL in 1–2 s; the 22 Oct glitches flagged	3 × 1 = 3
FM-18	Hardware	Zero offset of the sensor	Levels read 294 ppm high	1 × 3 = 3	The alarms use changes, not levels; the offset is corrected in the analysis	Outdoor zero, 4 Nov	1 × 2 = 2
FM-19	Hardware	Automatic self-calibration left at its factory setting (on)	After 24 h of power it can re-zero the sensor mid-procedure, and the step could trip the absence rule	2 × 1 = 2	No session ran for 24 h. Designed: switch it off at power-up (the driver has the command)	Found in review, not tested	2 × 1 = 2
FM-20	Environment	Condensation from humid breath in the line or cell	A blocked line (FM-2) or a damaged sensor	3 × 1 = 3	Tube end held away from the mouth, short line. Designed: a water trap in the line, as on the kit list	Not seen on the bench	3 × 1 = 3

Four of these failures changed the design. The sealed cell that held old breath led to the draw (D2); the impossible jumps of 22 Oct set the sensor-fault limits (FM-17); the frozen ring of 25 Oct moved the waiting into the sensor driver (FM-12); and the cell that read green at the ceiling for minutes on 1 Nov produced the top-of-scale rule (FM-5). The kit's non-rebreather mask, which delivered far less CO2 to the line than the bare tube end, was rejected as the sampling interface before the scored tests (D2) rather than mitigated.

SECTION 10, DELIVERABLE D9

Traceability, limitations and technical memorandum

Every clinical requirement traces to at least one ER, and every ER to a test, its evidence and a verdict. CR-1 is met for presence; CR-2 is met in 9 of 10 runs; CR-3 and CR-4 are partly met, each with its measured cause and a designed fix.

10.1 Traceability matrix

CR	ER	TEST	EVIDENCE	VERDICT
CR-1	ER-1 CO2 present within 3 breaths	5	`data/runs/test5_summary.csv`, results.csv; `data/raw/run_2026-11-04_*_t5-run*.log`	Pass
CR-1, CR-4	ER-3 response measured against line length	6	`data/runs/tube_length_results.csv`, `response_10_90.csv`	Pass
CR-1, CR-4	ER-4 range, accuracy and drift	1	`data/calibration/2026-11-04_outdoor_zero.md`; `data/raw/co2_2026-11-04_212406.log`	Partly met
CR-1, CR-2	ER-2 CRITICAL within 30 s; dashboard within 2 s; phones within 10 s	5, 7	Test 5 logs; the alarm and notification tables of `data/co2monitor.sqlite` (laptop)	Not met (9 of 10)
CR-3	ER-5 a disconnected line is a fault	2, 3	`data/raw/co2_2026-11-02_175529.log`, 18:00 to 18:13	Pass
CR-3	ER-6 a blocked line is a fault	4	`data/raw/co2_2026-11-03_003113.log`, 00:42	Not met
CR-4	ER-7 breathing rate from the CO2 signal	Paced run; 22 Oct replay	`data/raw/rate_2026-11-04_205550_rr-run01.log`; `data/runs/rr_replay_2026-10-22.csv`, `peak_rate_results.csv`, rate_results.csv	Not met
CR-4	ER-8 advisory on a sustained fall	3 Nov event; simulation	`data/raw/co2_2026-11-03_183654.log`; `tests/test_cr4.py`	Not verified on the bench

CR	STATUS
CR-1	Met for presence: CO2 shown present within 3 breaths in 10 of 10 runs (ER-1), with the response measured and published (ER-3). Absence was called within 30 s in 9 of 10 runs (ER-2, as for CR-2), and accuracy above the zero is not verified (ER-4)
CR-2	Met in 9 of 10 runs. The one late alarm (33 s) is explained by sensor saturation and has a designed fix, the fixed room-air dilution (ER-2)
CR-3	Partly met. A disconnected line is always a fault, never apnoea (ER-5); a blocked line still shows as apnoea, for want of pressure or flow sensing in the kit (ER-6)
CR-4	Partly met. The rate is recorded with every reading and was never falsely low, but it is resolved only when breaths reach the sensor directly (ER-7, not met); the advisory is verified only in simulation and was recorded once on the bench (ER-8, not verified)

10.2 Declared limitations

- **Bench only.** Breath came from team members at the open end of a line, with no patient circuit, lung simulator or reference capnograph; nothing here supports clinical use.
- **The values are diluted sample concentrations, not end-tidal.** Holding the tube end 10 cm from the mouth dilutes the breath about 10- to 45-fold so that it fits the 5,000 ppm sensor, and the dilution varied from run to run (peaks of 2,228 to 5,000 ppm). Neither the levels nor their changes map onto end-tidal CO2.
- **A blocked line is not detected** (ER-6). The kit has no means to sense line pressure or flow; the designed fix is pump-current sensing or a pressure sensor (FM-2).
- **The sensor sets the limits of CR-4 and of the timing.** Its 5,000 ppm ceiling is about a tenth of exhaled CO2 and its 10–90 % rise is 13.9 s, so no rate is resolved through the line, and a saturated cell sees a stop late (FM-5, FM-6).
- **The sample is drawn by hand.** The delay depends on the person pulling, and a stopped draw makes an apnoea look like a slow fall (FM-7, FM-9). The designed pump at a fixed flow replaces the syringe.
- **ER-1 was verified at 15 breaths/min with the 0.5 m line only.** A 2.1 m line adds about 16 s before CO2 shows (test 6).
- **Accuracy above the zero is not verified,** and zero drift is not watched while the box runs. The card's Figure 3 lists zero drift as a technical trigger; this device checks drift only offline (ER-4, FM-18, FM-19).
- **Small samples.** Ten runs per test give wide intervals: the sensitivity of 0.90 has a 95 % interval of 0.60 to 0.98.
- **The box runs from the laptop's USB port,** so a pulled cable silences it, and only the dashboard's "No data" alarm remains (FM-11).

TECHNICAL MEMORANDUM

What the CO2 monitor does, and how far it can be trusted

To: a non-specialist reader. **From:** EXSU 620 Team 6. **Subject:** what the CO2 monitor does, and how far it can be trusted.

The prototype checks whether a person is breathing out carbon dioxide (CO2). A team member pulls a small, steady sample of air with a syringe, through a thin tube from in front of the mouth into a sealed chamber, where an infrared sensor measures the CO2 once a second. When the level rises with breathing, a ring of lights on the box turns green; on the bench this happened by the second or third breath every time. When the level falls and stays down, as it would if breathing stopped or a breathing tube came out of the airway, the ring turns red and the box sounds an alarm: within 30 seconds in 9 of 10 bench tests, typically in 15. A laptop beside the box shows the same alarm on screen and sends a message to registered phones within a few seconds, although a sending limit once held one back. The box also turns amber when the CO2 falls slowly, as in hypoventilation, but that warning has been tried only in simulation and once on the bench. The box decides its alarms itself, so it keeps warning if the laptop program stops; it does depend on the laptop for power, so unplugging its USB cable turns it off.

If the sampling tube is pulled off, a clip on the tube tells the box, which shows a blue-green equipment warning instead of a breathing alarm. It cannot yet tell a blocked tube from a person who has stopped breathing: both turn it red. Its sensor is too slow to follow single breaths and reads only up to a tenth of the CO2 in a breath, so the sample is diluted with room air, and the numbers it shows are levels in that diluted sample, not the end-tidal values a clinical capnograph reports. It counts breaths only when they reach the sensor directly, and otherwise says "not resolved" rather than guess. In three of the ten tests the sensor reached its upper limit, and in one of them the alarm came at 33 seconds, 3 seconds past the target.

The next version, designed but not built, adds a small pump with a fixed dilution, which should keep the sample below the sensor's limit, and a way to sense a blocked tube; following single breaths would still need a faster sensor. This is a bench prototype tested on team members' breath, not a medical device.

STATEMENT

Use of generative AI

The team used Claude (Anthropic) throughout the project, as section 6.4 of the course handbook allows. It helped write and review the firmware, the laptop service, the dashboard, the tests and the scripts that generate the printed parts; it analysed the bench logs and drew the diagrams and charts (figures 1 to 5 and 8 to 11); it searched and summarised the literature; and it drafted and edited the text of this dossier. Team members built the hardware, ran every bench test and recorded the logs in `data/`. Every result in D7 comes from those saved logs, and each number and figure was checked against its log before submission. The team takes responsibility for all content.

SOURCES

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